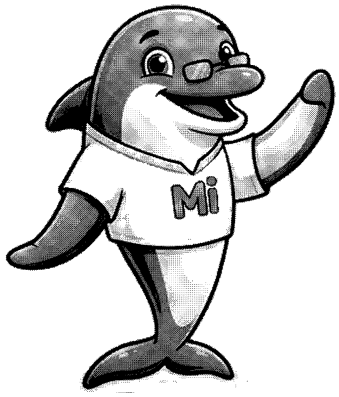
Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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Good to know

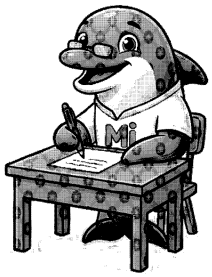
- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 79: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- The rulers in Unit 9 are printed at real size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

Contents

	About this book Copyright and licence	page 2
	Welcome	page 3
	My pace plan	page 5
1	**Number sleuths** Number sleuths (numbers to 10 000) · follows Holiday Program Lesson 1 · answers on page 80	page 6
2	**Cryptarithm crackers** Cryptarithm crackers · follows Holiday Program Lesson 2 · answers on page 82	page 13
3	**Times table treasure** Times table treasure · follows Holiday Program Lesson 3 · answers on page 85	page 20
4	**Model makers** Model makers (bar models) · follows Holiday Program Lesson 4 · answers on page 87	page 27
5	**Remainder riddles** Remainder riddles · follows Holiday Program Lesson 5 · answers on page 89	page 34
6	**Fraction walls** Fraction walls · follows Holiday Program Lesson 6 · answers on page 91	page 41
7	**Area architects** Area architects · follows Holiday Program Lesson 7 · answers on page 93	page 48
8	**Angle and line detectives** Angle and line detectives · follows Holiday Program Lesson 8 · answers on page 96	page 55
9	**Measure and money masters** Measure and money masters · follows Holiday Program Lesson 9 · answers on page 98	page 62
10	**Puzzle day** Puzzle day · follows Holiday Program Lesson 10 · answers on page 100	page 69
	My progress	page 76
	My certificate	page 77
	Answers Check your work, with short reasons	page 79

My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Number sleuths**	6-7	8-9	10 and 12	11	☐
2	**Cryptarithm crackers**	13-14	15-16	17 and 19	18	☐
3	**Times table treasure**	20-21	22-23	24 and 26	25	☐
4	**Model makers**	27-28	29-30	31 and 33	32	☐
5	**Remainder riddles**	34-35	36-37	38 and 40	39	☐
6	**Fraction walls**	41-42	43-44	45 and 47	46	☐
7	**Area architects**	48-49	50-51	52 and 54	53	☐
8	**Angle and line detectives**	55-56	57-58	59 and 61	60	☐
9	**Measure and money masters**	62-63	64-65	66 and 68	67	☐
10	**Puzzle day**	69-70	71-72	73 and 75	74	☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

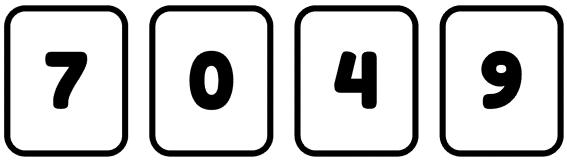
Learn it: Numbers to 10 000

My goal. I can read, build and order 4-digit numbers. I can use clues to find a mystery number.

A 4-digit number has thousands, hundreds, tens and ones. The place decides what a digit is worth. A 0 can never be the first digit.

Worked example

Use the cards 7, 0, 4 and 9 once each. Make the greatest and the smallest 4-digit number.



1. **Greatest: put the greatest digit in the thousands place. Then the next greatest. This gives 9740.**
2. **Smallest: the first digit cannot be 0. Use the smallest digit that is not 0: the 4.**
3. **Put the other digits in order after it, smallest first: 0, 7, 9. This gives 4079.**

Answer: Greatest 9740. Smallest 4079.

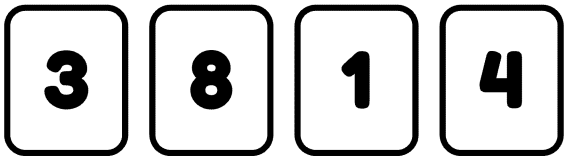
How to make a number from cards

1. Decide the thousands digit first.
2. Odd or even? Choose the ones digit first.
3. Making a list? Fix the first digit, then list the rest from smallest.

Your turn

Answer at the back of the book.

Try **Make the greatest even number with the cards 3, 8, 1 and 4. Use each card once.** ☐



Start here

Do these first. Take your time.

1 **Look at the place value chart.**

Th	H	T	O
4	0	7	2

(a) Write the number. ☐

(b) The digit 7 is worth ☐.

2 **Write each number in digits.**

(a) 6 thousands, 5 hundreds and 9 ones ☐

(b) 2 thousands and 15 tens ☐

(c) 5 thousands, 12 hundreds and 3 ones ☐

3 **Fill in the blanks.**

(a) 1000 more than 6457 is ☐.

(b) 100 less than 3050 is ☐.

(c) 10 more than 8995 is ☐.

(d) 1 less than 7000 is ☐.

4 **Arrange the numbers in order. Begin with the smallest.**
5830, 5083, 5380, 5038

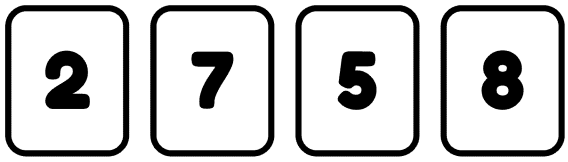
_____

5 **A shop has 3 boxes of 1000 pins, 14 packs of 100 pins and 6 loose pins. How many pins are there in all?** ☐

Start here

Keep going. Check your work.

6 **Use the cards once each in every number.**



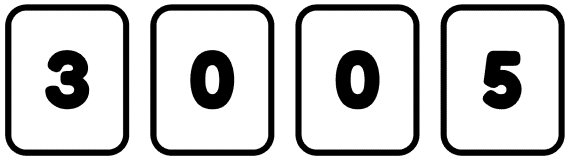
(a) The greatest number ☐

(b) The smallest number ☐

(c) The greatest odd number ☐

(d) The smallest odd number ☐

7 **List every 4-digit number that can be made with the cards 3, 0, 0 and 5. Begin with the smallest.**



_____

8 **Write the next 2 numbers in each pattern.**

(a) 2300, 2600, 2900, ☐, ☐

(b) 9000, 8750, 8500, ☐, ☐

(c) 3, 4, 6, 9, 13, ☐, ☐

9 **A pattern starts at 4000 and goes up by 500 each time.**

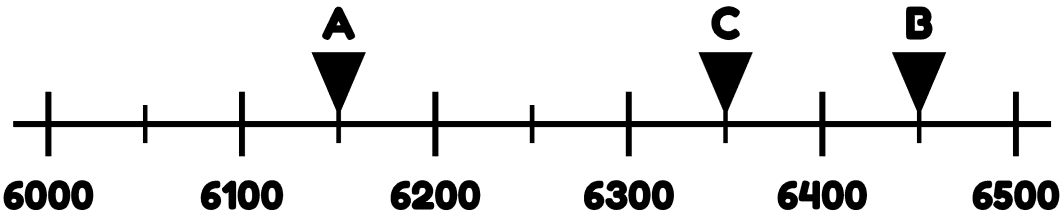
(a) Is 6800 in the pattern? Yes No

(b) The first number above 6800 in the pattern is ☐.

Start here

Nearly there. You can do it.

10 **Look at the number line.**



(a) Arrow A points to ☐.

(b) Arrow B points to ☐.

(c) Arrow C points to ☐.

11 **I am a 4-digit number. My thousands digit is 6, my hundreds digit is 3 less than that, and my tens digit is 0. My ones digit is the greatest odd digit.**

What number am I? ☐

12 **Ravi says the greatest odd number from the cards 9, 4, 6 and 2 is 9642.**

9	4	6	2

(a) Is he right? Yes No

(b) The greatest odd number from the cards is ☐.

13 **Look at the number 5737.**

(a) The first 7 is worth ☐.

(b) The second 7 is worth ☐.

(c) The first 7 is worth ☐ more than the second 7.

Think harder

Hints are upside down. Try first!

1 **The digits of a 4-digit number are all different and add up to 8. The number is even.**

(a) The greatest it can be is ☐.

(b) The smallest it can be is ☐.

Sets of digits: _____

Numbers I tried: _____

_____

_____

Hint. Find the sets of 4 different digits that add up to 8 first.

2 **Ali writes every number from 1 to 200. How many times does he write the digit 7?** ☐

1 to 99: _____

100 to 200: _____

Total: _____

_____

Hint. Count 1 to 99 first. Then count 100 to 200.

3 **Mei uses only the digits 1 and 2 to make 4-digit numbers. A digit can be used more than once. She writes them in a list, smallest first. Which number is 10th in her list?** ☐

My list: _____

_____

_____

Hint. Begin with 1111. Change the ones digit first. Keep a list.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Counting digit sums.**
Count the 3-digit numbers with a digit sum of 4. List them by first digit.

(a) Digit sum 4: ☐ numbers.

(b) Use the pattern. Digit sum 5: ☐ numbers.

First digit 4: _____

First digit 3: _____

Hint. Fix the first digit. See how the counts grow: 1, 2, 3, ...

2 **Beyond Level 3: Growing jumps.**
Look at the jumps between the numbers. Write the next 2 numbers.

5, 8, 14, 23, 35, ☐, ☐

Jumps: _____

Hint. Write the jumps between the numbers. What do you see?

3 **Make up your own number riddle. Choose a secret number with 4 digits, then write 3 clues. Test your clues: exactly ONE number must fit. Swap riddles with a grown-up.**

Clue 1: _____

Clue 2: _____

Clue 3: _____

Hint. Pick the number first. Then write clues about its digits.

4 **Find the secret number. It is greater than 3000 and less than 4000. Its digits are all different and add up to 15. The tens digit is 0, and the hundreds digit is double the ones digit.** ☐

Hint. Try ones digit 1, 2, 3 and 4. Check each sum.

✔ Check yourself

No hints. Try each one on your own.

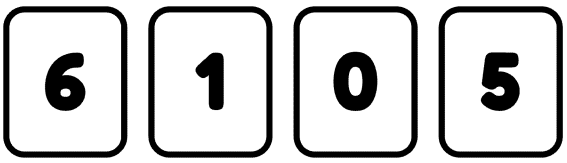
1 In 7258, the digit 2 is worth ☐.

Working: _____

2 Write the next 2 numbers. 6500, 6000, 5500, ☐, ☐

Working: _____

3 Make the smallest 4-digit number from the cards 6, 1, 0 and 5. Use each card once. ☐



4 Ali has 5 boxes of 1000 beads, 12 bags of 100 beads and 8 loose beads. How many beads does he have? ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Addition and subtraction

My goal. I can find missing digits in sums and take-aways. I can solve magic squares and number pyramids.

In a missing-digit sum, work one column at a time. Start with the ones. Remember to add the digit you carried.

Worked example

Find the 2 missing digits in this sum.

	Th	H	T	O
	4	5	□	7
+	1	□	6	8
	5	8	9	5

1. Ones: 7 + 8 = 15. Write 5 and carry 1.
2. Tens: ? + 6 + 1 carried must end in 9. So ? is 2.
3. Hundreds: 5 + ? makes 8. So ? is 3.
4. Check the thousands: 4 + 1 = 5. It matches.

Answer: 4527 + 1368 = 5895. The missing digits are 2 and 3.

How to find a missing digit

1. Look at the ones column first.
2. Write the carried 1 above the next column.
3. For a take-away, turn it round and add to check.

Your turn

Answer at the back of the book.

Try Find the missing digit. □

3?48 + 2517 = 5965

Start here

Do these first. Take your time.

1 **Find the missing digit in each sum.**

(a) 3?46 + 1251 = 4597 The missing digit is ☐.

(b) 26?9 + 1435 = 4104 The missing digit is ☐.

2 **Find the missing digit in each take-away.**

(a) 6405 − 2?18 = 3687 The missing digit is ☐.

(b) 9?30 − 4617 = 4513 The missing digit is ☐.

3 **A theatre sells 3875 tickets on Saturday and 2968 tickets on Sunday.**

(a) It sells ☐ tickets in all.

(b) It sells ☐ more tickets on Saturday than on Sunday.

4 **A baker has 8000 g of flour. He uses 2475 g on Monday and 3150 g on Tuesday. How many grams of flour are left?** ☐ g

5 **Siti works out 5000 − 1768 = 3342.**

(a) Check by adding: 3342 + 1768 = ☐

(b) Is Siti right? Yes No

(c) The correct answer is ☐.

6 **Look for a pattern.**

(a) 2641 + ☐ = 9999

(b) 10 000 − 2641 = ☐

Start here

Keep going. Check your work.

7 **Every row, column and diagonal has the same total.**

(a) The magic total is ☐.

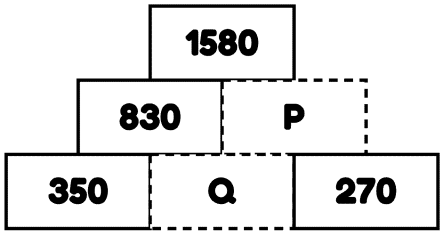
(b) A = ☐ B = ☐ C = ☐

200	450	A
550	350	150
300	B	C

8 **A brick is made by adding the 2 bricks below it.**

(a) P = ☐

(b) Q = ☐



9 **Find the 2 missing digits.**

(a) The digit in the hundreds place of the top number is ☐.

(b) The digit in the tens place of the number being added is ☐.

	Th	H	T	O
	3	☐	5	2
+	2	4	☐	6
	6	1	4	8

10 **Each letter stands for a digit. The same letter always means the same digit. Find the digit.**

(a) A3 + 2A = 56 A = ☐

(b) C5 + C5 = 90 C = ☐

Start here

Nearly there. You can do it.

11 **Find the 2 missing digits.**

(a) The digit in the hundreds place of the top number is ☐.

(b) The digit in the tens place of the number taken away is ☐.

	Th	H	T	O
	6	☐	0	2
−	2	8	☐	5
	3	6	5	7

12 **The sum of 2 numbers is 9000. One number is 3475. What is the other number?** ☐

Working: _____

13 **Use the cards once each to make two 3-digit numbers.**

2	3	4	5	7	8

(a) The greatest sum is ☐.

(b) The smallest sum is ☐.

14 **A school collects cans for 3 weeks: 1875, 2430 and 1790.**

(a) In all, the school collects ☐ cans.

(b) Week 2 has ☐ more cans than Week 3.

Working: _____

Think harder

Hints are upside down. Try first!

1 **Letters hide digits. The same letter is the same digit. Different letters never share a digit. Find the digits.**

AB + AC = BBD

(a) A = ☐ **(b)** B = ☐ **(c)** C = ☐ **(d)** D = ☐

My tries: _____

_____

Hint. Two 2-digit numbers add up to less than 200. What is B?

2 **Use the cards 1, 3, 5, 6 and 8 once each. Make a 3-digit number and a 2-digit number. What is the smallest sum you can make?** ☐

1	3	5	6	8

My tries: _____

_____

_____

Hint. Which digits are worth the least? Which places are worth most?

3 **Look at this magic square. Every row, column and diagonal has the same total.**

120	270	A
	210	
180		E

(a) The magic total is ☐.

(b) A = ☐

(c) E = ☐

Hint. The middle number is in 4 lines. Add those 4 lines.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Letters hide digits. The same letter is the same digit. Different letters never share a digit. Find the digits.**

ABC + BC = CAB

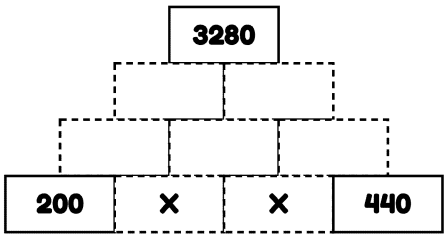
(a) A = ☐ **(b)** B = ☐ **(c)** C = ☐

My tries: _____

Hint. Look at the ones: C + C ends in B. List the choices for C.

2 **Beyond Level 3: Pyramids with a hidden number.**

A brick is made by adding the 2 bricks below it. The 2 bricks marked X are equal. Find X. ☐



Hint. Guess a number for X. Fill the pyramid. Is the top too big?

3 **Beyond Level 3: Reversing digits.**

Reverse the digits of a 2-digit number and add the 2 numbers.

(a) 36 + 63 = ☐ **(b)** 58 + 85 = ☐ **(c)** 47 + 74 = ☐

(d) Each answer is in the ☐ times table.

Hint. Add the tens digit and ones digit. Compare with the answers.

4 **Make up a number pyramid puzzle of your own. The bottom row has 3 bricks. Show exactly 3 bricks and hide the other 3, so that only ONE answer fits. Swap puzzles with a grown-up.**

Draw my pyramid:

Hint. Make a full pyramid first. Then hide 3 bricks.

✔ Check yourself

No hints. Try each one on your own.

1 **Find the missing digit.** ☐

4?72 + 2519 = 7091

Working: _____

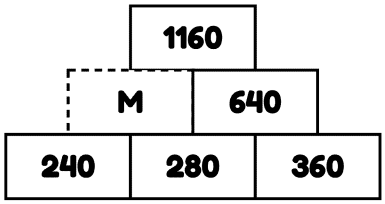
2 **Find the missing digit.** ☐

8?04 - 3569 = 4635

Working: _____

3 **A brick is made by adding the 2 bricks below it.**

The brick M is ☐ **.**



4 **A zoo has 9000 visitors in 3 days. On Friday 2845 visitors come. On Saturday 3260 come. How many come on Sunday?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

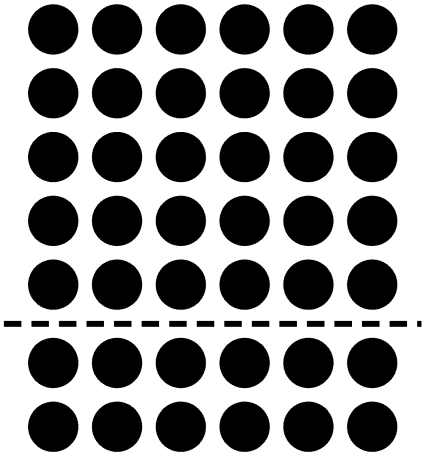
Learn it: Tables of 6, 7, 8 and 9

My goal. I can use facts I know to find new times table facts. I can find patterns and solve number puzzles.

A times table fact you do not know can be built from facts you do know. Split a rectangle of dots into two smaller rectangles.

Worked example

There are 7 rows with 6 dots in each row. How many dots are there in all?



1. **The dashed line splits the dots into 5 rows and 2 rows.**
2. **5 × 6 = 30 and 2 × 6 = 12.**
3. **Add the 2 parts: 30 + 12 = 42.**
4. **So 7 × 6 = 42.**

Answer: 7 × 6 = 42. There are 42 dots.

Ways to find a fact

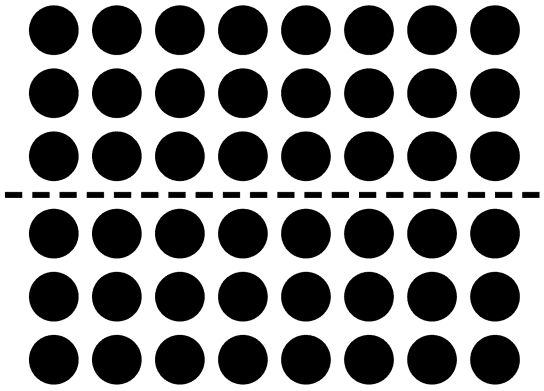
1. Split into facts you know: 5 ×, 2 ×, 10 ×.
2. For 9 ×: take 1 group away from 10 groups.
3. Check with the last digit.

Your turn

Answer at the back of the book.

Try

There are 6 rows with 8 dots in each row. Split them into 3 rows and 3 rows. How many dots are there in all?



Start here

Do these first. Take your time.

1 **Write the missing answers in the 8 times table.**

n	1	2	3	4	5	6	7	8	9	10
8 × n	8	16	24		40		56	64		80

2 **Write the answers.**

(a) 6 × 8 = ☐ (b) 7 × 9 = ☐ (c) 9 × 6 = ☐

(d) 8 × 7 = ☐

3 **You know that 8 × 5 = 40. Use it to find the other facts.**

(a) 8 × 6 = ☐

(b) 8 × 4 = ☐

(c) 8 × 10 = ☐

(d) 8 × 9 = ☐

4 **Ali packs 7 eggs in each box. He fills 9 boxes.**

(a) He packs ☐ eggs.

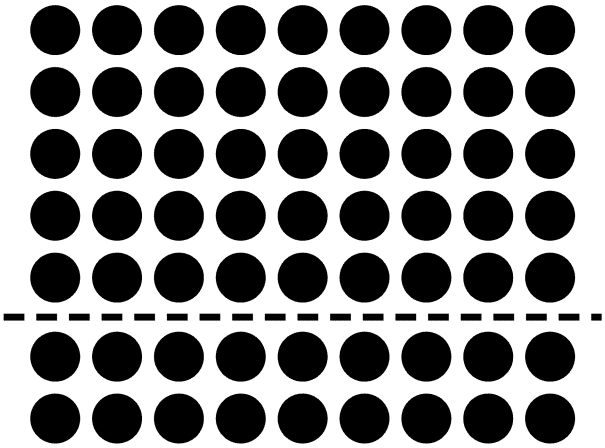
(b) Then he packs 56 eggs into boxes of 7. He fills ☐ boxes.

5 **Find my number. It is in the 7 times table, and it is greater than 50 and less than 80. Its digits add up to 9.** ☐

Start here

Keep going. Check your work.

6 **The dashed line splits 7 rows of 9 dots into 2 parts.**



(a) The top part has 5 × 9 = ☐ dots.

(b) The bottom part has 2 × 9 = ☐ dots.

(c) There are ☐ dots in all.

7 **Fill in the missing answers.**

×	6	7	8	9
6	36	42	A	54
7	B	49	56	C
8	48	D	64	72

A = ☐ B = ☐ C = ☐ D = ☐

8 **A row number times a column number makes each product. Write the missing numbers in the heading cells.**

×		
	54	63
	48	56

Start here

Nearly there. You can do it.

9 **Write the last digit of each answer. Do not find the whole answer.**

(a) 6 × 13 ☐

(b) 8 × 27 ☐

(c) 9 × 45 ☐

(d) 7 × 68 ☐

10 **Mei puts 30 tiles in equal rows. There are at least 2 rows and at least 2 tiles in each row. List ALL the ways.**

There are ☐ ways.

rows	tiles in each row

11 **Which of these numbers are in the 6 times table?**
24, 38, 42, 56, 60

_____

12 **Mei buys 6 packs of 8 stickers. Ben buys 7 packs of 7 stickers.**

(a) Who has more stickers? Mei Ben

(b) How many more? ☐

Think harder

Hints are upside down. Try first!

1 **Ben writes the 7 times table from 7 × 1 to 7 × 30. How many of his answers end in 4?** ☐

The 7 times table: _____

_____

The answers ending in 4: _____

_____

Hint. Which number times 7 ends in 4? Then find the next ones.

2 **The row numbers and column numbers are all from 6 to 9. Find the empty cells A to D. You can write in the heading cells to help you.**

(a) A = ☐ (b) B = ☐

(c) C = ☐ (d) D = ☐

×			
	42	48	A
	49	B	63
	C	D	72

Hint. Start with 49. Which row and column make 49?

3 **Two numbers have a product of 63 and a sum of 16. What are the 2 numbers?** ☐ **and** ☐

Pairs that multiply to 63: _____

_____

_____

Hint. List the pairs of numbers that multiply to make 63.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Digit sums and the 9s.**
Add the digits again and again: 684 gives 6 + 8 + 4 = 18, then 1 + 8 = 9.
Numbers in the 9 times table always end at 9. Which of these are in it?
738, 825, 963, 1089

_____

Hint. Add the digits of each number. Keep adding until 1 digit is left.

2 **Beyond Level 3: Adding every row.**
A hall has 8 rows. Row 1 has 7 seats. Each row has 2 more seats than the row before.

Row	1	2	3	4	5	6	7	8
Seats	7	9						

(a) Row 8 has ☐ seats.

(b) The hall has ☐ seats in all.

Hint. Pair the first and last rows. Then the next pair.

3 **Make your own 2 by 2 grid puzzle with numbers from 6 to 9. Show 3 products. Only ONE answer may fit. Swap it with a grown-up.**

_____

Hint. Pick the row and column numbers first. Then multiply.

4 **Ben multiplies a number by 8.**

(a) His answer ends in 6. The last digit of his number could be ☐ or ☐.

(b) His answer ends in 5. The last digit of his number could be

0 2 5 7 none .

Hint. List the last digits of 8 × 1 up to 8 × 10.

✔ Check yourself

No hints. Try each one on your own.

1 **A box holds 8 pens. How many pens are in 7 boxes?** ☐

Working: _____

2 **You know that 5 × 7 = 35. Use it to find 6 × 7.** ☐

Working: _____

3 **What is the last digit of 7 × 56?** ☐

Working: _____

4 **Mei has 54 apples. She puts 6 apples in each bag. How many bags does she need?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

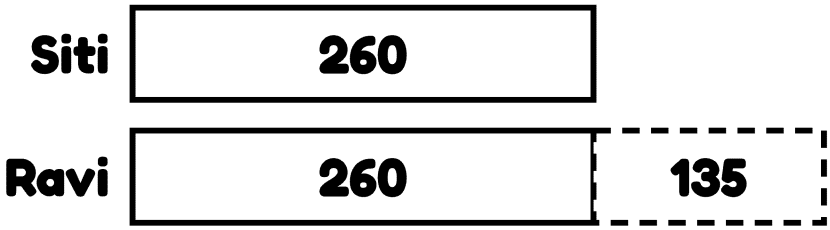
Learn it: Word problems with bar models

My goal. I can draw a bar model for a word problem. I can use it to decide what to add, subtract or divide.

A bar model is a picture of a word problem. Each bar is a length that shows an amount. Longer bar, greater amount.

Worked example

Siti has 260 beads. Ravi has 135 more beads than Siti. How many beads do they have in all?



1. **Draw a bar for Siti. Draw a longer bar for Ravi. The extra part is 135.**
2. **Ravi has 260 + 135 = 395 beads.**
3. **Both children: 260 + 395 = 655 beads.**

Answer: They have 655 beads in all.

How to draw a bar model

1. Draw one bar for each amount.
2. Write what you know on the bars. Put a ? on what you want to find.
3. Look at the bars. Then add, subtract or divide.

Your turn

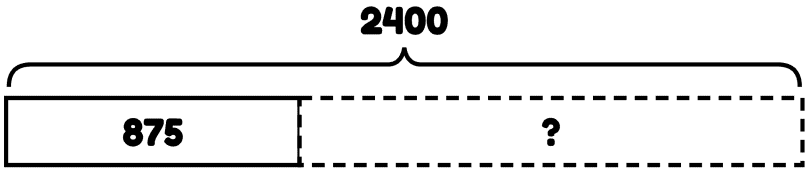
Answer at the back of the book.

Try **Ali has 480 stamps. Ben has 215 fewer stamps than Ali. How many stamps does Ben have?** ☐

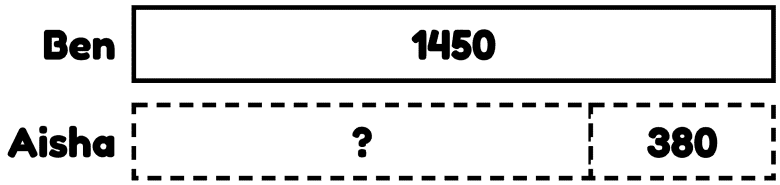
Start here

Do these first. Take your time.

1 A farm has 2400 eggs. 875 of them are brown. The rest are white. How many eggs are white? ☐



2 Ben has 1450 cards. Aisha has 380 fewer cards than Ben.



(a) Aisha has ☐ cards.

(b) They have ☐ cards in all.

3 A school sells 468 tickets on Friday, 539 on Saturday and 392 on Sunday.

(a) It sells ☐ tickets in all.

(b) It sells ☐ more tickets on Saturday than on Sunday.

4 Siti has 64 beads. Wei has 29 beads. Each child gets 12 more beads.

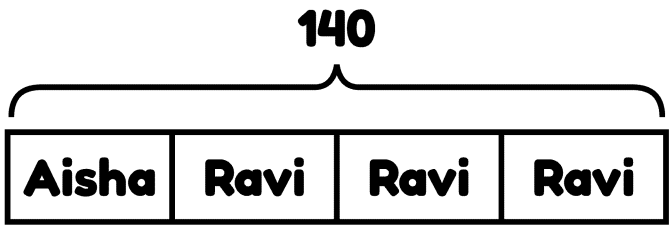
(a) Before, Siti has ☐ more beads than Wei.

(b) After, Siti has ☐ more beads than Wei.

Start here

Keep going. Check your work.

5 **Ravi and Aisha have 140 coins in all. Ravi has 3 times as many coins as Aisha.**



(a) Aisha has ☐ coins.

(b) Ravi has ☐ coins.

Working: _____

6 **Kumar and Wei have 90 stickers in all. Kumar has 18 more stickers than Wei.**

(a) Wei has ☐ stickers.

(b) Kumar has ☐ stickers.

Draw your model:

Working: _____

7 **Ravi's ribbon is 78 cm long. Mei's ribbon is 52 cm long. Ravi cuts a piece off his ribbon and gives it to Mei. Now both ribbons are the same length. How long is the piece?** ☐ **cm**

Draw your model:

Working: _____

Start here

Nearly there. You can do it.

8 **Ravi reads 85 pages on Monday. On Tuesday he reads 28 pages more than he does on Monday. On Wednesday he reads 17 pages fewer than on Tuesday.**

(a) He reads ☐ pages on Tuesday.

(b) He reads ☐ pages on Wednesday.

(c) He reads ☐ pages in the 3 days.

9 **Mei has $50. She buys a book for $18.40 and a pen for $6.75. How much money does she have left? $** ☐

10 **Wei has 3 times as many marbles as Aisha. Wei has 48 marbles.**

(a) Aisha has ☐ marbles.

(b) They have ☐ marbles in all.

Draw your model:

11 **Ali has 54 stickers. Ben has 18 fewer stickers than Ali. Ravi says Ben has 54 + 18 = 72 stickers.**

(a) Is Ravi right? Yes No

(b) Ben has ☐ stickers.

12 **Siti saves $12 each week for 7 weeks. Then she spends $35. How much money does she have left? $** ☐

Working: _____

Think harder

Hints are upside down. Try first!

1 **Kumar and Wei have 84 marbles in all. Kumar has 3 times as many marbles as Wei. Kumar gives some marbles to Wei. After that, each boy has the same number of marbles. How many marbles does Kumar give away?** ☐

Draw your model:

Working: _____

Hint. Find how many each has at first. Then how many each has later.

2 **Ali buys a pencil box and 2 pens for $14. Mei buys the same box and 5 pens for $23.**

(a) 1 pen costs $ ☐ .

(b) The pencil box costs $ ☐ .

Draw your model:

Working: _____

_____

Hint. Compare the 2 bills. What do the 3 extra pens cost?

3 **Ravi is 7 years old. His father is 35 years old. In how many years will his father be twice as old as Ravi?** ☐ **years**

Draw your model:

Working: _____

Hint. The age gap never changes. Draw a bar for each age.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Aisha, Kumar and Lily have 140 stickers in all. Kumar has 12 more stickers than Lily. Aisha has twice as many stickers as Kumar.**

(a) Lily has ☐ stickers. **(b)** Kumar has ☐ stickers.

(c) Aisha has ☐ stickers.

Draw your model:

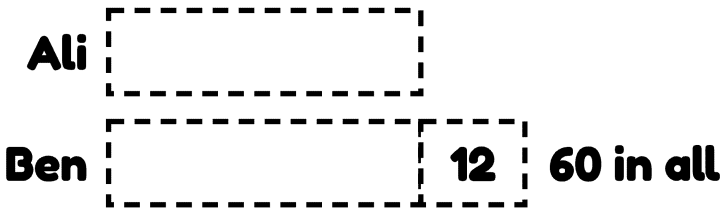
Hint. Draw Lily as 1 bar. Build Kumar and Aisha from that bar.

2 **Beyond Level 3: One amount changes.**
A shop has 4 times as many red balloons as blue balloons. It sells 18 red balloons. Now the shop has the same number of red and blue balloons. How many balloons did the shop have at first? ☐

Draw your model:

Hint. Draw blue as 1 box and red as 4 boxes. Selling 18 leaves 1 box.

3 **Look at the bar model. Make up a story that fits it. Then solve your story.**



My story: _____

_____

My answer: _____

Hint. Pick things that fit: stickers, coins, cards or beads.

✔ Check yourself

No hints. Try each one on your own.

1 **A library has 2000 books. 735 books are lent out. 640 books are being mended. How many books are on the shelves?** ☐

Working: _____

2 **Ravi has 560 beads. Siti has 135 more beads than Ravi. How many beads do they have in all?** ☐

Working: _____

3 **Siti and Kumar have 72 stamps in all. Siti has 5 times the number of stamps that Kumar has. How many stamps does Siti have?** ☐

Working: _____

4 **Two boxes hold 50 pens in all. Box A has 8 more pens than Box B. How many pens are in Box A?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
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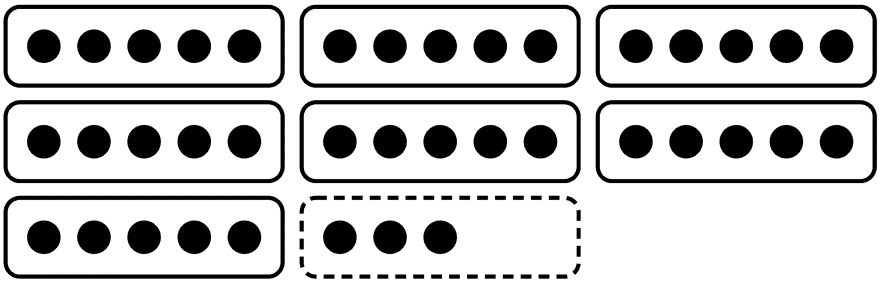
Learn it: Division with remainders

My goal. I can divide with a remainder. I can decide what the remainder means in a story. I can find a mystery number.

When we share or make groups, some may be left over. The leftover is the remainder. It is always smaller than the number we divide by.

Worked example

Ali has 38 stickers. Each bag takes 5 stickers. How many bags can he fill? How many stickers are left over?



1. **Find the greatest fact that is not more than 38. 5 × 7 = 35.**
2. **Subtract to find what is left: 38 − 35 = 3.**
3. **Write it as 38 ÷ 5 = 7 remainder 3.**
4. **Check: 7 × 5 + 3 = 38.**

Answer: 7 full bags, with 3 stickers left over.

How to divide with a remainder

1. Find the greatest table fact that fits in the number.
2. Subtract it to find the remainder.
3. Check: groups × size + remainder = start.

Your turn

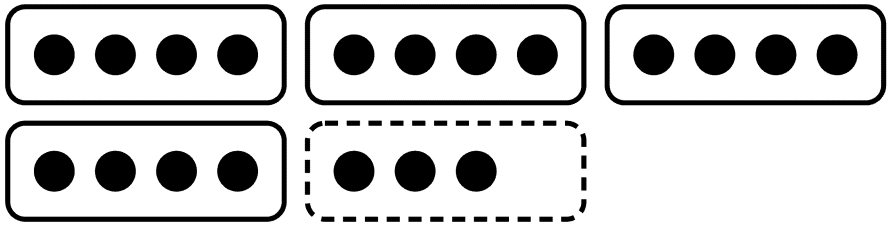
Answer at the back of the book.

Try **Mei has 43 beads. She makes bracelets with 6 beads each. She makes** ☐ **full bracelets and has** ☐ **beads left over.**

Start here

Do these first. Take your time.

1 **The dots are put into groups of 4.**



(a) There are ☐ full groups. (b) There are ☐ dots left over.

2 **Divide.**

(a) 38 ÷ 7 = ☐ remainder ☐

(b) 50 ÷ 8 = ☐ remainder ☐

(c) 61 ÷ 9 = ☐ remainder ☐

(d) 75 ÷ 6 = ☐ remainder ☐

3 **Divide. Then decide what the leftover means.**

(a) 26 people go in cars of 5. They need ☐ cars.

(b) 37 cm of ribbon, cut in 6 cm pieces, gives ☐ whole pieces.

(c) Ravi has $30. A toy costs $4. He buys ☐ toys.

(d) Then he has $ ☐ left.

4 **Kumar says 59 ÷ 7 = 7 remainder 10.**

(a) The remainder 10 is greater than ☐.

(b) The right answer is ☐ remainder ☐.

Start here

Keep going. Check your work.

5 **Ravi puts marbles into bags of 6. He fills 9 bags and has 4 marbles left over.**

(a) Ravi has ☐ marbles.

(b) He packs all the marbles again into bags of 7. He fills ☐ bags and has ☐ marbles left.

6 **Think about remainders.**

(a) List every remainder that is possible when a number is divided by 5.

_____

(b) A number between 70 and 80 is divided by 9. The remainder is as great as it can be. The number is ☐.

7 **Look for a pattern. Write the remainder when each number is divided by 4.**

Number	12	13	14	15	16	17	18	19
Remainder								

(a) Fill in the empty cells.

(b) When 30 is divided by 4 the remainder is ☐.

(c) When 100 is divided by 4 the remainder is ☐.

8 **Find a number between 30 and 40. Dividing it by 6 gives remainder 1. Dividing it by 4 gives remainder 3.**

(a) The numbers that leave remainder 1 when divided by 6 are ☐ and ☐.

(b) The number I am looking for is ☐.

Start here

Nearly there. You can do it.

9 **52 children go on a trip. Each bus has 9 seats.**

(a) The children need ☐ buses.

(b) The last bus has ☐ children.

(c) ☐ seats on the last bus are empty.

10 **The letters A, B, C and D repeat in this order.**

Place	1	2	3	4	5	6	7	8
Letter	A	B	C	D	A	B	C	D

(a) The 30th letter is ☐.

(b) The 32nd letter is ☐.

11 **Mei has 8 bags with 9 beads in each bag, and 5 loose beads. She shares all the beads equally among 9 friends.**

(a) Mei has ☐ beads in all.

(b) Each friend gets ☐ beads and ☐ beads are left over.

12 **Circle each number that has remainder 3 when you divide it by 5.**

18 23 33 41 48

13 **Siti has 53 stickers. She puts 8 stickers on each card.**

(a) She makes ☐ full cards.

(b) She needs ☐ more stickers to make one more full card.

Think harder

Hints are upside down. Try first!

1 **Today is Tuesday.**

(a) 45 days from today it will be

Mon Tue Wed Thu Fri Sat Sun

(b) 20 days ago it was

Mon Tue Wed Thu Fri Sat Sun

Working: _____

_____

Hint. A week has 7 days. Divide by 7 and use the remainder.

2 **Find the 2-digit number. Dividing it by 5 gives remainder 2. Dividing it by 3 gives remainder 1. Dividing it by 4 gives remainder 3.** ☐

My list: _____

Tests: _____

_____

Hint. Begin with the clue that gives the fewest numbers to list.

3 **Aisha shares crayons among her cousins. If each cousin gets 3 crayons, 11 crayons are not used. If each cousin gets 5 crayons, she needs 7 more. How many cousins are there? How many crayons does she have?**

(a) There are ☐ cousins.

(b) She has ☐ crayons.

Working: _____

_____

Hint. Compare the 2 plans: each cousin gets 2 more crayons.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Remainders when you add and double.**
A number leaves remainder 3 when divided by 7.

(a) Add 5 to it. The new remainder is ☐.

(b) Double it. The new remainder is ☐.

(c) Another number leaves 5 when divided by 8. Doubled, it leaves ☐.

My test numbers: _____

Hint. Try a number like 10 (remainder 3 when divided by 7). Test it.

2 **Ali has between 20 and 100 marbles. Counting in 3s leaves 1 over. Counting in 4s leaves 2 over. Counting in 5s leaves 3 over. How many marbles does he have?** ☐

Hint. It is always 2 less than a table number.

3 **Make up your own number riddle: a range and 2 remainder clues. Only ONE number may fit. Swap with a grown-up.**

My riddle: _____

The numbers I tested: _____

Hint. Pick your number first. Then find its remainders.

4 **Beyond Level 3: Making numbers with 2 sizes.**
Ben sells eggs in boxes of 3 and boxes of 5. An order must be whole boxes.

(a) Can he fill an order for 11 eggs? Yes No

(b) Orders from 1 to 15 he CANNOT fill: ☐ ☐ ☐ ☐

Hint. List the orders he can fill. Which numbers are missing?

✔ Check yourself

No hints. Try each one on your own.

1 **Divide. 52 ÷ 8 = ☐ remainder ☐**

Working: _____

2 **A bus holds 9 people. How many buses are needed for 40 people? ☐**

Working: _____

3 **Mei packs 25 muffins into boxes of 6. How many muffins are left over?** ☐

Working: _____

4 **Find a number between 20 and 30. It leaves remainder 2 when divided by 3 and remainder 4 when divided by 5.** ☐

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
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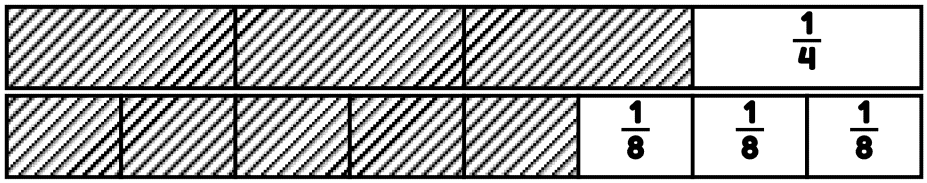
Learn it: Fractions

My goal. I can use a fraction wall to find equal fractions. I can compare fractions and add and subtract them.

A fraction wall shows one whole cut into equal parts. A longer piece of the wall is a greater fraction. To compare fractions, look at where the pieces end.

Worked example 1

Which is greater, 5/8 or 3/4? Use the fraction wall.

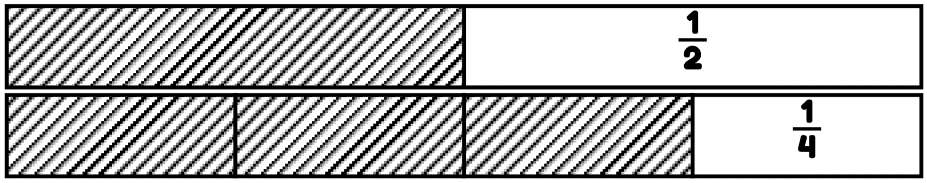


1. Shade 5/8 on the row of eighths.
2. Shade 3/4 on the row of quarters.
3. The end of 3/4 lines up with the end of 6/8.
4. 6/8 is more than 5/8, so 3/4 is greater.

Answer: 3/4 is greater than 5/8.

Worked example 2

What is 1/2 + 1/4? Use the fraction wall.



1. On the wall, 1/2 is the same length as 2/4.
2. Add the quarters: 2/4 + 1/4 = 3/4.

Answer: 1/2 + 1/4 = 3/4.

How to compare and add fractions

1. Make the parts the same size. Use the wall.
2. To add or subtract, change both to the same bottom number.
3. Add or subtract the top numbers. Keep the bottom number.

Your turn

Answer at the back of the book.

Try Circle <, = or >. 7/12 < = > 2/3

Start here

Do these first. Take your time.

1 Use the fraction wall to fill in the blanks.

$\frac{1}{3}$		$\frac{1}{3}$		$\frac{1}{3}$	
$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$

(a) 1/3 = ☐ /6 = ☐ /12

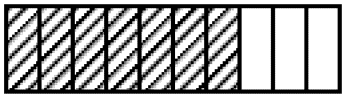
(b) 2/3 = ☐ /6 = ☐ /12

(c) 5/6 = ☐ /12

2 Write the fraction of each shape that is shaded.

A 

B 

C 

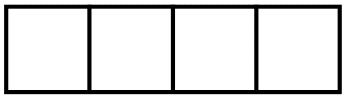
D 

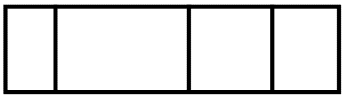
(a) A = ☐ / ☐ (b) B = ☐ / ☐

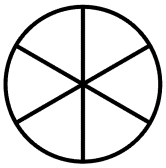
(c) C = ☐ / ☐ (d) D = ☐ / ☐

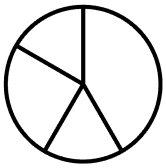
3 Which shapes are cut into EQUAL parts? Circle the letters.

A B C D

A 

B 

C 

D 

Name _____

Start here

Keep going. Check your work.

4 Circle <, = or > in each part.

(a) 5/8 < = > 3/8

(b) 1/3 < = > 1/4

(c) 2/4 < = > 4/8

(d) 3/4 < = > 5/6

5 Arrange the fractions in order. Begin with the smallest.
5/6, 1/3, 1/2, 2/3

_____

6 Fill in the blanks.

(a) 3/10 + 4/10 = ☐ /10

(b) 5/8 - 2/8 = ☐ /8

(c) 9/12 - 4/12 = ☐ /12

(d) 1 - 3/10 = ☐ /10

7 Change to the same bottom number. Then fill in the blanks.

(a) 1/2 + 1/4 = ☐ /4

(b) 1/3 + 1/6 = ☐ /6

(c) 3/4 - 1/8 = ☐ /8

(d) 5/6 - 1/2 = ☐ /6

Start here

Nearly there. You can do it.

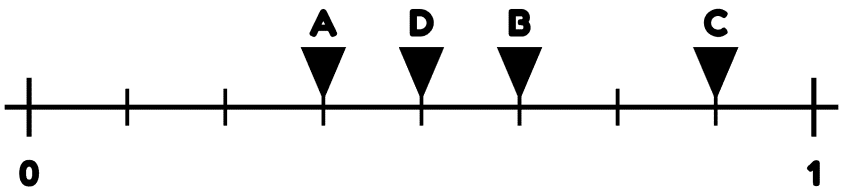
8 **A cake is cut into 12 equal slices. Ali eats 3 slices. Ben eats 4 slices.**

(a) Ali and Ben eat ☐/12 of the cake.

(b) ☐/12 of the cake is left.

(c) Ben eats ☐/12 more of the cake than Ali.

9 **Look at the number line from 0 to 1. It is cut into equal steps.**



(a) A = ☐/8

(b) B = ☐/8

(c) C = ☐/8

(d) Which arrow shows 1/2? A B C D

10 **Fill in the blanks to make equal fractions.**

(a) 3/4 = ☐/12 (b) 2/5 = ☐/10 (c) 1/2 = 5/☐

(d) 6/8 = 3/☐

11 **Siti says 3/8 + 2/8 = 5/16.**

(a) Is she right? Yes No

(b) The right answer is ☐/8.

Think harder

Hints are upside down. Try first!

1 **Look at the fractions with bottom number 12. List every one between 1/3 and 2/3.**

_____

Working: _____

_____

_____

_____

Hint. Change 1/3 and 2/3 into twelfths first.

2 **I am a fraction equal to 2/5. My bottom number is 9 more than my top number. Which fraction am I?** ☐ / ☐

My list: _____

_____

_____

Hint. Write fractions equal to 2/5. Look at the gap each time.

3 **A jug is full of juice. Mei drinks 1/4 of the jug. Ali drinks 1/3 of the jug.**

(a) Together they drink ☐ /12 of the jug.

(b) ☐ /12 of the jug is left.

Working: _____

_____

_____

Hint. Change quarters and thirds into twelfths first.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Adding fractions with top number 1.**
Write each fraction as a sum of 2 fractions that each have top number 1.

(a) 3/4 = 1/2 + 1/ ☐ **(b)** 5/6 = 1/2 + 1/ ☐

(c) 7/12 = 1/3 + 1/ ☐

Hint. Take away the first fraction. What is left over?

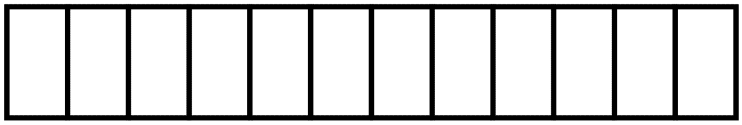
2 **A whole bar is filled with exactly 6 pieces. The pieces are 1/2, 1/4 or 1/8 of the bar. How many of each?**

(a) ☐ halves, ☐ quarters and ☐ eighths.

My tries: _____

Hint. Eighths are the smallest. Most of the 6 pieces must be small.

3 **Colour the bar. Draw lines in 1/2 of the bar. Draw dots in 1/3 of the bar. Leave the rest white.**



(a) ☐ parts are white.

(b) That is the same as 1/ ☐ of the bar.

Hint. 1/2 of 12 parts is 6 parts. How many parts for 1/3?

4 **Make up a fraction story about a pizza cut into 8 equal slices. The answer to your story must be 5/8.**

My story: _____

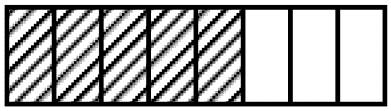
My answer: _____

Hint. Pick 2 fractions with bottom number 8 that add up to 5/8.

✔ Check yourself

No hints. Try each one on your own.

1 Write the fraction that is shaded.



☐ /8

2 Circle <, = or >. 5/8 < = > 3/4

Working: _____

3 5/8 − 1/4 = ☐ /8

Working: _____

4 Fill in the blank. 2/3 = ☐ /12

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

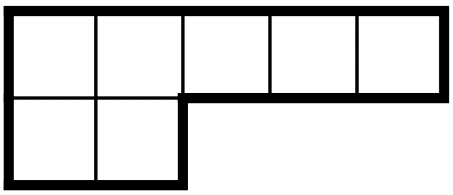
Learn it: Area and perimeter

My goal. I can find the area and the perimeter of rectangles and of shapes made of squares.

Area is the flat space inside a shape. We count squares and write cm². Perimeter is the distance all the way round the edge, in cm.

Worked example

The figure is made of squares with sides of 1 cm. What are its area and its perimeter?



1. **Area: count the squares. 5 + 2 = 7, so the area is 7 cm².**
2. **Perimeter: add the sides all the way round. 5 + 1 + 3 + 1 + 2 + 2 = 14 cm.**
3. **Check: the figure fits in a box 5 cm by 2 cm. 2 × (5 + 2) = 14 cm.**

Answer: Area 7 cm². Perimeter 14 cm.

How to find area and perimeter

1. Area: count the squares, or length × width.
2. Perimeter: add all the sides.
3. Rectangle: perimeter = 2 × (length + width).
4. Write cm for the perimeter and cm² for the area.

Your turn

Answer at the back of the book.

Try **A rectangle is 8 cm long and 3 cm wide.**

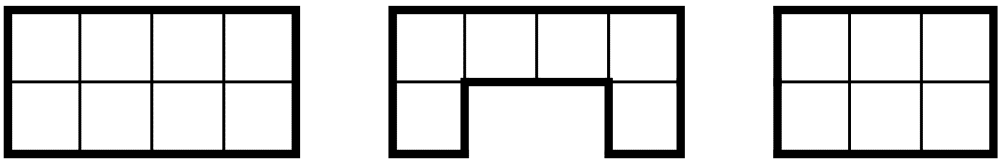
Its area is ☐ **cm². Its perimeter is** ☐ **cm.**



Start here

Do these first. Take your time.

1 **Every square in the picture has sides of 1 cm. What are the area and the perimeter of each figure?**



A B C

(a) A: area ☐ cm², perimeter ☐ cm

(b) B: area ☐ cm², perimeter ☐ cm

(c) C: area ☐ cm², perimeter ☐ cm

(d) Same area, different perimeter: A,B A,C B,C

2 **What are the area and the perimeter of each rectangle?**

(a) 8 cm long and 6 cm wide: area ☐ cm², perimeter ☐ cm

(b) 11 m long and 4 m wide: area ☐ m², perimeter ☐ m

3 **Look at squares.**

(a) A square has sides of 7 cm. Its perimeter is ☐ cm.

(b) Its area is ☐ cm².

(c) A square garden has a fence 36 m long all round. Each side is ☐ m.

4 **A rectangle has a perimeter of 30 cm. Its length is 9 cm.**

(a) Its width is ☐ cm. (b) Its area is ☐ cm².

Start here

Keep going. Check your work.

5 **A mat is 15 cm long and 9 cm wide. Square tiles with sides of 3 cm cover it exactly.**

(a) ☐ tiles fit along the length.

(b) ☐ tiles fit along the width.

(c) The mat needs ☐ tiles in all.

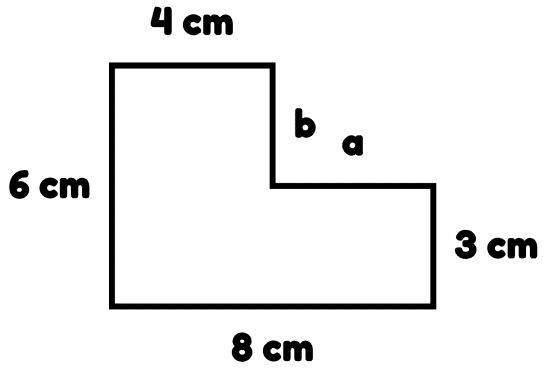
6 **Mei has 14 cm of wire. She bends all of it into a rectangle. All the sides are whole numbers of cm. List ALL the rectangles.**

(a) There are ☐ different rectangles.

(b) The one with the greatest area has an area of ☐ cm².

length	width	area

7 **Find the missing sides a and b first. Then find the area, and then the perimeter.**



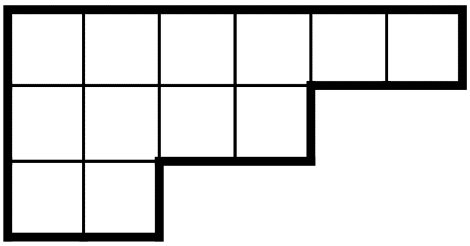
(a) a = ☐ cm (b) b = ☐ cm (c) The area is ☐ cm².

(d) The perimeter is ☐ cm.

Start here

Nearly there. You can do it.

8 **Every square in the picture has sides of 1 cm.**



(a) The area is ☐ cm². (b) The perimeter is ☐ cm.

(c) Ben fills the gaps to make a 6 by 3 rectangle. He adds ☐ squares.

(d) The perimeter of the rectangle is ☐ cm.

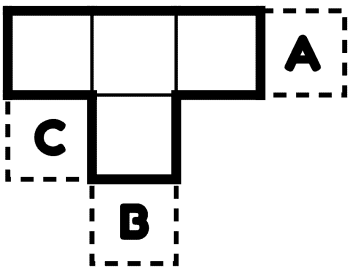
9 **Mei draws a rectangle 10 cm long and 2 cm wide. Ben draws a square with the same perimeter.**

(a) Each side of Ben's square is ☐ cm.

(b) Whose shape has the greater area? Mei's Ben's

(c) It is greater by ☐ cm².

10 **The figure has a perimeter of 10 cm. Ali adds ONE square at A, B or C.**



(a) Which places make the perimeter 12 cm? A B C

(b) Which place keeps the perimeter 10 cm?
A B C

11 **A square room has a floor area of 64 m².**

(a) Each side of the room is ☐ m long.

(b) A border goes all round the floor. It is ☐ m long.

Think harder

Hints are upside down. Try first!

1 **A square has sides of 8 cm. One straight cut makes it into 2 equal rectangles.**

(a) Each rectangle has a perimeter of ☐ cm.

(b) The 2 rectangles together have a perimeter of ☐ cm.

(c) That is ☐ cm more than the perimeter of the square.

Working: _____

Hint. Each rectangle is 8 cm by 4 cm. Draw it.

2 **A rectangle has area 36 cm² and perimeter 26 cm.**

(a) Its length is ☐ cm.

(b) Its width is ☐ cm.

Pairs that multiply to 36: _____

_____

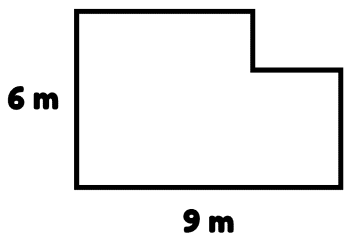
Hint. List pairs that multiply to 36. Then add each pair.

3 **A rectangle is 9 m long and 6 m wide. A rectangle 3 m by 2 m is cut out of one corner, as shown.**

(a) The area that is left is ☐ m².

(b) The perimeter of the figure is ☐ m.

(c) The same piece is cut from the MIDDLE of a long side. The perimeter is then ☐ m.



Hint. Push the corner out. Then think about a notch in the middle.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 3: Perimeter patterns.

Squares with sides of 1 cm are joined in a row. 1 square has a perimeter of 4 cm. 2 squares have 6 cm. 3 squares have 8 cm.

(a) A row of 10 squares has a perimeter of ☐ cm.

(b) A row with a perimeter of 30 cm has ☐ squares.

My pattern: _____

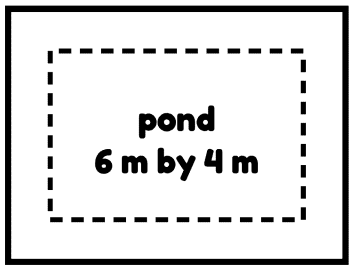
Hint. The perimeter goes up by 2 cm for each new square.

2 Beyond Level 3: A path round a pond.

A pond is 6 m long and 4 m wide. A path 1 m wide goes all round the pond.

(a) Pond and path make a rectangle ☐ m by ☐ m.

(b) The area of the path is ☐ m^2.



Hint. Find the area of the whole rectangle. Take away the pond.

3 Beyond Level 3: The largest square tile.

A floor is 12 m long and 10 m wide. It is covered exactly with square tiles of one size. The side of a tile is a whole number of metres.

(a) The largest tile has sides of ☐ m.

(b) The floor needs ☐ of these tiles.

My list: _____

Hint. The tile side must fit exactly in 12 and in 10.

✔ Check yourself

No hints. Try each one on your own.

1 **A rectangle is 9 cm long and 5 cm wide. Its area is ☐ cm².**

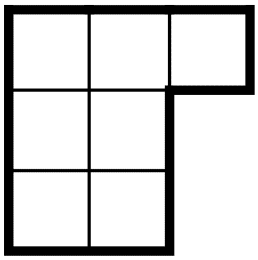
Working: _____

2 **A square has sides of 6 cm. Its perimeter is ☐ cm.**

Working: _____

3

Every square has sides of 1 cm. The area is ☐ cm². The perimeter is ☐ cm.



4 **A rectangle has a perimeter of 22 cm. Its width is 4 cm. Its length is ☐ cm.**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

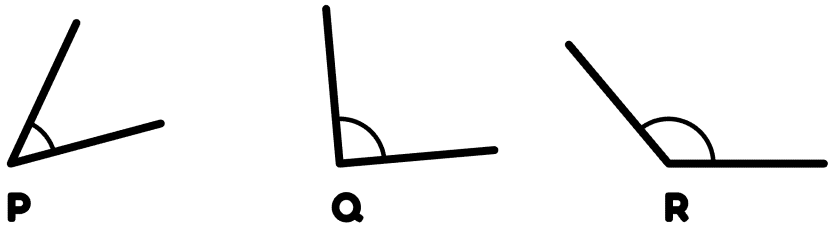
Learn it: Angles and lines

My goal. I can compare angles with a right angle. I can find perpendicular and parallel lines. I can count angles in a figure.

An angle is made where 2 lines meet. A right angle is a square corner, like the corner of a page. Use the corner of a piece of paper as a right angle checker.

Worked example

Use a right angle checker. Is each angle smaller than, equal to, or greater than a right angle?



1. Put the paper corner on the point where the lines meet.
2. Line up one side of the paper with one line.
3. Look at the other line. If it is inside the corner, the angle is smaller. If it is outside, the angle is greater.

Answer: P is smaller than a right angle. Q is a right angle. R is greater than a right angle.

Lines to remember

1. Perpendicular lines meet at a right angle.
2. Parallel lines never meet. They stay the same distance apart.
3. Count angles in order: the small ones first.

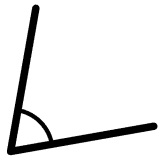
Your turn

Answer at the back of the book.

Try

Use a right angle checker. This angle is

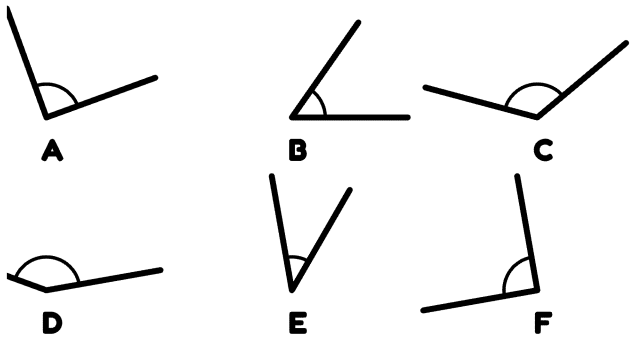
smaller than equal to greater than a right angle.



Start here

Do these first. Take your time.

1 **Use a right angle checker on each angle. Write the letters.**

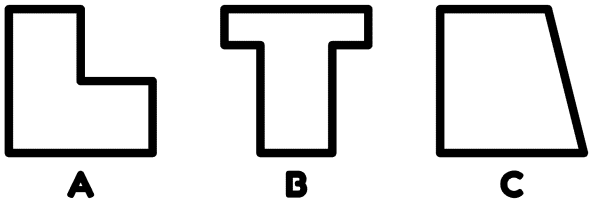


(a) Smaller than a right angle: _____

(b) A right angle: _____

(c) Greater than a right angle: _____

2 **Count the right angles INSIDE each shape. Check each corner. A dent is greater than a right angle, so leave it out.**



(a) A: ☐ right angles (b) B: ☐ right angles

(c) C: ☐ right angles

3 **Look at the letters.**

F V N L

(a) Circle each letter with 2 PARALLEL lines. F V N L

(b) Circle each letter with 2 PERPENDICULAR lines. F V N L

Start here

Keep going. Check your work.

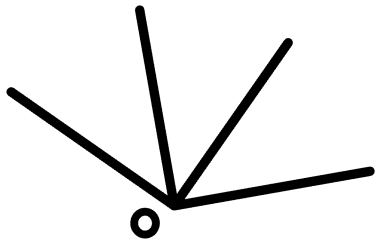
4 **These 4 lines all start at point O. Count the angles. Count the small ones first.**

(a) Angles made of 1 small angle: ☐

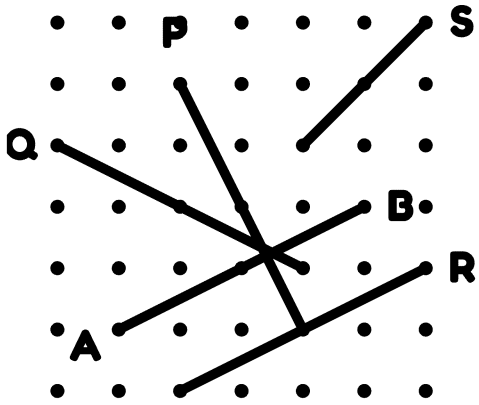
(b) Made of 2 small angles: ☐

(c) Made of 3 small angles: ☐

(d) Angles in all: ☐



5 **Look at the lines on the dot grid.**



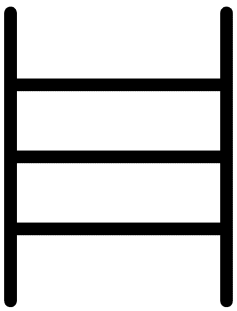
(a) Which line is PARALLEL to line AB? P Q R S

(b) Which line is PERPENDICULAR to line AB? P Q R S

6 **A ladder has 2 upright rails and 3 rungs. Each rung meets a rail at right angles.**

(a) Where one rung meets the 2 rails there are ☐ right angles.

(b) The ladder has ☐ right angles in all.



Start here

Nearly there. You can do it.

7 **Mei faces north. She makes turns of a right angle.**

(a) She turns 1 right angle to the right. She faces N E S W

(b) From north, she turns 3 right angles to the right. She faces
N E S W

(c) From north, she turns 2 right angles to the left. She faces
N E S W

8 **Fill in the blanks.**

(a) A rectangle has ☐ right angles.

(b) A rectangle has ☐ pairs of parallel sides.

(c) A triangle has at most ☐ right angle.

(d) Two sides that meet at a corner of a square are
parallel perpendicular

9 **Circle True or False.**

(a) Lines that are perpendicular cross at a right angle. True False

(b) Parallel lines meet if you make them long enough. True False

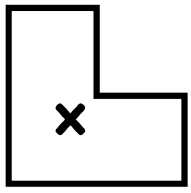
(c) One line can be perpendicular to 2 parallel lines. True False

(d) Two lines that cross are always perpendicular. True False

10 **Siti says angle X is a right angle.**

(a) Is she right? Yes No

(b) Angle X is
smaller than | equal to | greater than
a right angle.



Think harder

Hints are upside down. Try first!

1 **Lines from a point O make angles. 2 lines give 1 angle, 3 lines give 3 angles and 4 lines give 6 angles.**

(a) With 5 lines there are ☐ angles.

(b) With 7 lines there are ☐ angles.

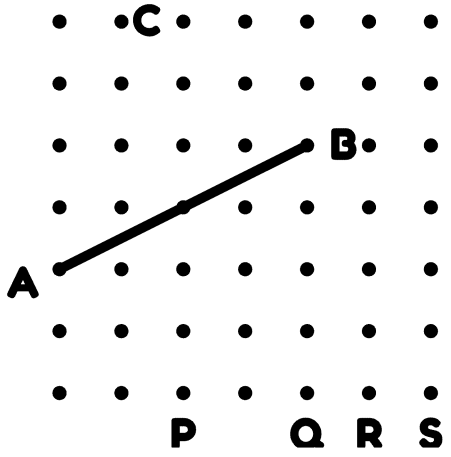
Working: _____

Hint. Small angles first: 4. Then angles made of 2 small ones: 3.

2 **Draw a line from C that is PERPENDICULAR to line AB. Use a ruler. Make it reach the bottom row of dots.**

(a) Your line reaches the bottom row at

P Q R S



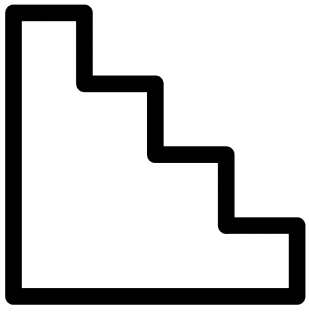
Hint. AB goes 2 across and 1 up. Turn it a quarter turn.

3 **A staircase has 4 steps.**

(a) It has ☐ right angles inside.

(b) It has ☐ dents.

(c) A staircase with 8 steps has ☐ right angles inside.



Hint. Count the dents first. Each new step adds 1 right angle.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Degrees.**
A right angle is 90 degrees.

(a) 2 right angles make a straight line: ☐ degrees.

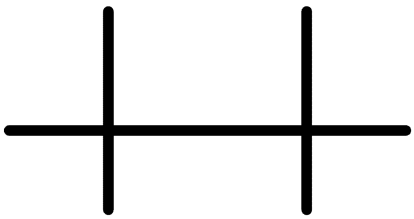
(b) 4 right angles make a full turn: ☐ degrees.

(c) Half of a right angle is ☐ degrees.

(d) 3 right angles make ☐ degrees.

Hint. Add the right angles: 90 + 90 + …

2 **Beyond Level 3: Lines and a third line.**
Two upright lines both cross the long line at right angles.



(a) The 2 upright lines are parallel perpendicular

(b) A line is perpendicular to one of 2 parallel lines. Is it also perpendicular to the other? Yes No

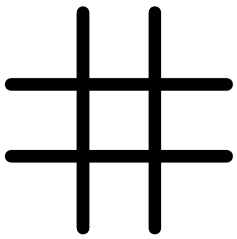
Hint. Do the 2 upright lines ever meet?

3 **A board has 2 lines across and 2 lines up.**

(a) There are ☐ pairs of parallel lines.

(b) The lines make ☐ right angles where they cross.

(c) A board with 3 lines across and 3 lines up makes ☐ right angles.



Hint. Each place where 2 lines cross makes 4 right angles.

✔ Check yourself

No hints. Try each one on your own.

1

Use a right angle checker. This angle is

smaller than equal to greater than a right angle.



2 **Circle each letter with 2 parallel lines.** **E H V T**

E H V T

3 **A rectangle has ☐ pairs of parallel sides.**

Working: _____

4 **Ben faces east. He turns 1 right angle to the left. He faces**

N E S W

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

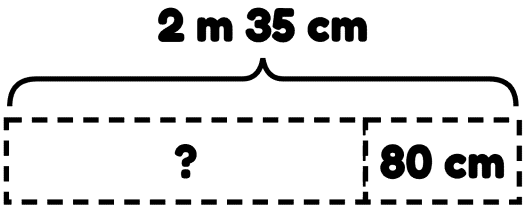
Learn it: Length, mass, volume, money and time

My goal. I can change units, solve money problems and find how long things last.

Use one unit before you add or take away. 1 m = 100 cm, 1 kg = 1000 g, 1 ℓ = 1000 ml and $1 = 100¢.

Worked example 1

A ribbon is 2 m 35 cm long. Mei cuts off 80 cm. How long is the ribbon now?

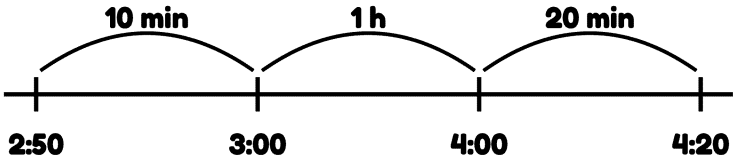


1. Change to cm: 2 m 35 cm = 235 cm.
2. Take away: 235 − 80 = 155 cm = 1 m 55 cm.

Answer: The ribbon is 1 m 55 cm long.

Worked example 2

A film starts at 2:50 pm and ends at 4:20 pm. How long is the film?



1. Jump to the next o'clock, then to 4:00, then to 4:20.
2. Add the jumps: 10 min + 1 h + 20 min = 1 h 30 min.

Answer: The film is 1 h 30 min long.

Tips

1. Write both amounts in the small unit first.
2. On a time line, jump to the next o'clock.

Your turn

Answer at the back of the book.

Try **A jug holds 2 ℓ 350 ml. Ali pours out 800 ml. The jug now holds** ☐ ℓ ☐ **ml.**

Start here

Do these first. Take your time.

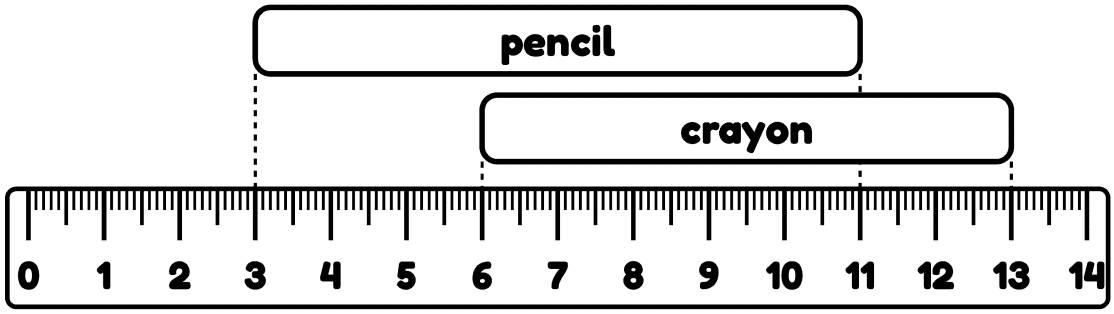
1 **Change the units.**

(a) 3 m = ☐ cm (b) 4 kg 50 g = ☐ g

(c) 5 ℓ 300 ml = ☐ ml (d) 2750 g = ☐ kg ☐ g

(e) 365 cm = ☐ m ☐ cm

2 **The ruler is shown at real size. Read the length of each object.**



(a) The pencil is ☐ cm long.

(b) The crayon is ☐ cm long.

(c) The pencil is ☐ cm longer than the crayon.

3 **The parcels weigh: A 1 kg 80 g, B 1900 g, C 1 kg 8 g, D 1 kg 90 g. Write the letters in order. Begin with the lightest.**

☐ ☐ ☐ ☐

4 **A jug has 2 ℓ 500 ml of juice. Ali pours it into cups of 400 ml.**

(a) He fills ☐ cups.

(b) ☐ ml of juice are left over.

Start here

Keep going. Check your work.

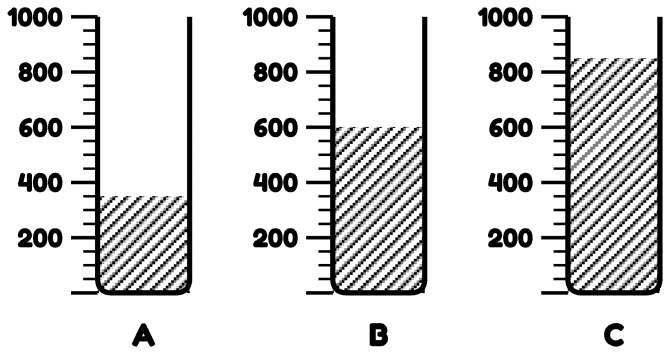
5 **Read the water level in each jug.**

(a) A holds ☐ ml

(b) B holds ☐ ml

(c) C holds ☐ ml

(d) C has ☐ ml more than A



6 **A notebook costs $2.65 and a ruler costs $1.40. Mei pays with a $5 note.**

(a) They cost $ ☐ in all.

(b) Her change is $ ☐.

7 **Siti has $20 and buys the first 3 things on the list.**

Book	Puzzle	Bottle	Stickers
$6.20	$4.85	$3.40	$1.75

(a) They cost $ ☐ in all.

(b) Her change is $ ☐.

(c) Can she buy all 4 things on the list? Yes No

8 **A box with 6 books weighs 2 kg 250 g. The empty box weighs 450 g. All the books have the same mass. One book weighs ☐ g.**

Working: _____

Start here

Nearly there. You can do it.

9 **Write the time in the other kind of time.**

(a) 6:15 am = ☐ (b) 1:40 pm = ☐ (c) 9:05 pm = ☐

(d) 16:50 = ☐ pm (e) 20:25 = ☐ pm

10 **How long does each take? Use a time line to help.**

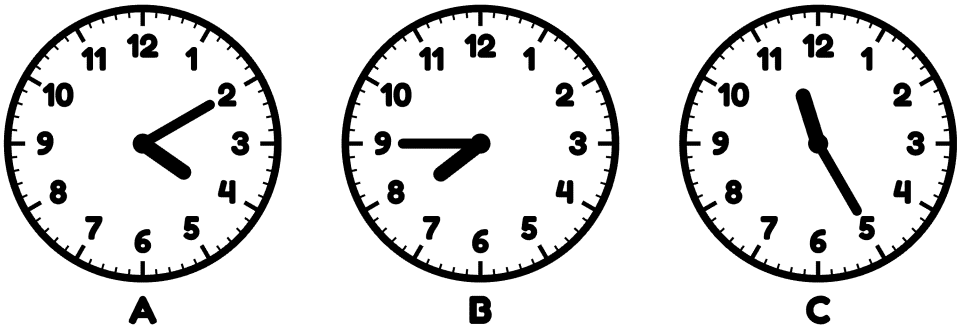
(a) Aisha's class: 5:25 pm to 6:40 pm. It lasts ☐ h ☐ min.

(b) Ben's class starts at 3:40 pm and lasts 55 minutes. It ends at ☐.

(c) A show ends at 8:15 pm. It is 1 h 45 min long. It starts at ☐.

Time line: _____

11 **Write the time shown on each clock.**



(a) A: ☐ (b) B: ☐ (c) C: ☐

12 **A stamp costs 65¢. Kumar buys 8 stamps.**

(a) The stamps cost $ ☐.

(b) He pays with a $10 note. His change is $ ☐.

Think harder

Hints are upside down. Try first!

1 **The buses leave the interchange every 20 minutes, starting at 06:50. The ride takes 25 minutes. Mei must reach school by 08:15. When does the latest bus she can take leave?** ☐

Bus times: 06:50, _____

_____

_____

Hint. Find the latest leaving time first: 08:15 minus 25 minutes.

2 **A night bus leaves at 22:50 and arrives at 02:10 the next day. How long is the trip?** ☐ **h** ☐ **min**

Time line: _____

_____

_____

Hint. Jump to midnight first. Then jump to 02:10.

3 **2 notebooks and 3 pens cost $9. A notebook costs $1.50 more than a pen.**

(a) A pen costs $ ☐ .

(b) A notebook costs $ ☐ .

Working: _____

_____

_____

Hint. Take away the extra $1.50 for each notebook first.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: A water puzzle.**
You have a 3 ℓ jug, a 5 ℓ jug and a tap. The jugs have no marks. You want exactly 4 ℓ.

(a) Fill the 5 ℓ jug. Pour from it into the 3 ℓ jug until full. The 5 ℓ jug has ☐ ℓ left.

(b) Empty the 3 ℓ jug. Pour the 2 ℓ into it. Fill the 5 ℓ jug. Pour from it into the 3 ℓ jug until full. The 5 ℓ jug has ☐ ℓ left.

Hint. Follow each step. The jug is full when it holds 3 ℓ.

2 **Mei's watch is 8 minutes slow.**

(a) Her watch shows 4:20 pm. The real time is ☐.

(b) A film starts at 4:45 pm. When it starts, her watch shows ☐.

Hint. A slow watch shows an earlier time than the real time.

3 **Write a shopping story that uses $6.40 and $2.15. The answer to your story must be $4.25.**

My story: _____

_____

My answer: _____

Hint. Which sum or take-away of $6.40 and $2.15 gives $4.25?

4 **A film lasts 1 h 55 min. It ends at 00:20, just after midnight. At what time did it start? Write it in 24-hour time.** ☐

Time line: _____

Hint. Go back to midnight first. Then go back the rest.

✔ Check yourself

No hints. Try each one on your own.

1 **2 m 40 cm = ☐ cm**

Working: _____

2 **A bottle holds 1 ℓ 200 ml. A child drinks 700 ml. The bottle then holds ☐ ml.**

Working: _____

3 **A film starts at 6:45 pm and ends at 8:10 pm. It lasts ☐ h ☐ min.**

Working: _____

4 **Kumar buys a snack for $2.35. He pays with a $5 note. His change is $ ☐ .**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

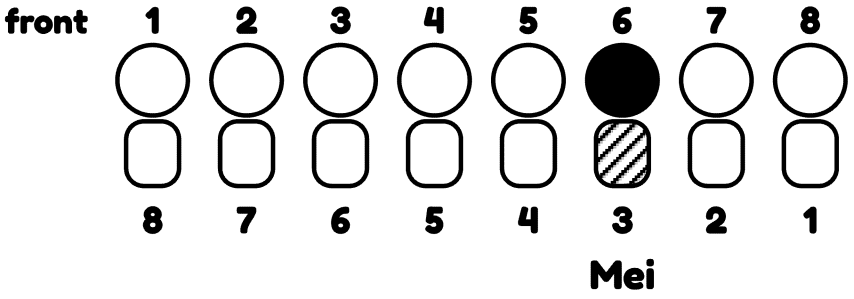
Learn it: Mixed puzzles

My goal. I can start a puzzle by drawing it, listing it or making a table. I can read a bar graph and solve logic puzzles.

To start a puzzle, try one of 3 ways. Draw it, list it, or make a table. Then check every clue.

Worked example

Mei is 6th from the front and 3rd from the back of a line. Count the children in the whole line.



1. **Draw the line. Put Mei in 6th place from the front.**
2. **Mei is 3rd from the back, so 2 children are behind her.**
3. **5 children are in front of Mei, then Mei, then 2 behind: 5 + 1 + 2 = 8.**
4. **Check: 6 + 3 − 1 = 8.**

Answer: There are 8 children in the line.

Three ways to start

1. Draw it: make a picture of the puzzle.
2. List it: write every case in order.
3. Table: write what you know in rows and columns.
4. Check your answer against every clue.

Your turn

Answer at the back of the book.

Try **Siti is 2nd from the front and 7th from the back of a line. The line has ☐ children.**

Start here

Do these first. Take your time.

1 **The bar graph shows the books 4 children read. Kumar read 15 more books than Siti. Draw Kumar's bar on the graph.**

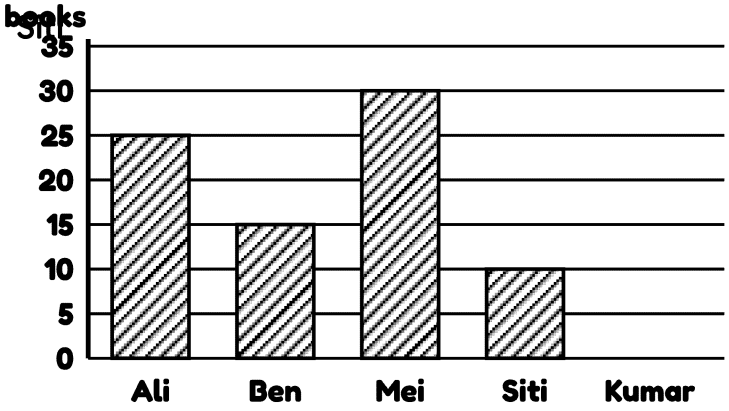
(a) Who read the most books?

Ali Ben Mei Siti

(b) Mei read ☐ more than Ben.

(c) Kumar read ☐.

(d) In all they read ☐.



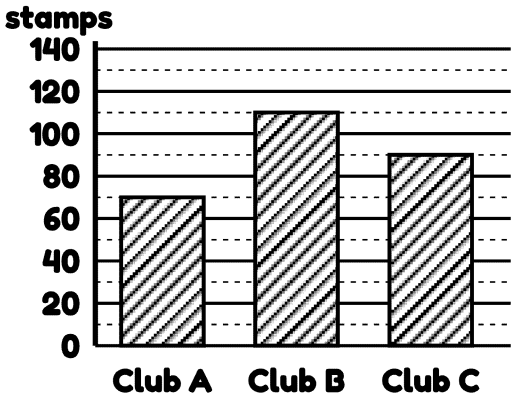
2 **The graph shows the stamps 3 clubs collected. Each thin line is 10 stamps.**

(a) Club A: ☐

(b) Club B: ☐

(c) Club C: ☐

(d) All 3: ☐



3 **The 3rd of a month is a Tuesday.**

(a) The 24th is a Mon Tue Wed Thu Fri Sat Sun

(b) The 30th is a Mon Tue Wed Thu Fri Sat Sun

Working: _____

Start here

Keep going. Check your work.

4 **Each child has a different fruit. Kumar has neither the apple nor the banana. The child with the orange has a name with 4 letters. Wei does not have the apple. Use ticks and crosses in the grid.**

	apple	banana	grapes	orange
Kumar				
Aisha				
Wei				
Lily				

(a) Kumar has the apple banana grapes orange

(b) Aisha has the apple banana grapes orange

(c) Wei has the apple banana grapes orange

(d) Lily has the apple banana grapes orange

5 **Draw it. Count the children in each line.**

(a) Ravi is 3rd from the front. Mei is last. 5 children stand between them. The line has ☐ children.

(b) Aisha is 8th from the front and 2nd from the back. The line has ☐ children.

Draw it:

6 **Make every 3-digit number from the cards 3, 6 and 9. Use each card once in a number. List them from the smallest.**

(a) _____

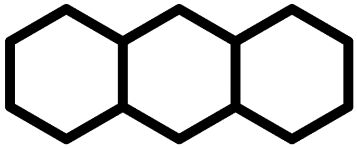
(b) There are ☐ even numbers in your list.

(c) The number closest to 700 is ☐.

Start here

Nearly there. You can do it.

7 **Hexagons are joined in a row with sticks. 1 hexagon needs 6 sticks. 2 hexagons need 11 sticks. 3 hexagons need 16 sticks. Make a table.**



(a) 6 hexagons need ☐ sticks.

(b) 41 sticks make ☐ hexagons.

hexagons	sticks

8 **Which numbers fit all the clues? The number is even and has 3 digits. It is less than 400. Its digits add up to 9, and none is 0. The tens digit is more than the ones digit.**

_____

9 **Two numbers add up to 60. The greater number is 14 more than the smaller one. Find the 2 numbers.** ☐ **and** ☐

Guess, check, improve: _____

10 **Aisha has $10. A juice costs $1.85 and a sandwich costs $3.40. She buys 2 juices and 1 sandwich. Her change is $** ☐ **.**

Think harder

Hints are upside down. Try first!

1 **7 friends meet. Each friend shakes hands once with every other friend. How many handshakes are there?** ☐

friends	handshakes

Hint. Try 2, 3, 4 friends first. Make a table.

2 **One of Ben, Aisha and Wei broke a cup. They each say 1 thing: Ben "Wei did it", Aisha "Not me", Wei "Ben is lying". Exactly 1 of them tells the truth. Who broke the cup? Ben Aisha Wei**

If Ben did it: _____

If Aisha did it: _____

If Wei did it: _____

Hint. Test each child as the one who broke the cup.

3 **A graph shows the pencils sold on 4 days. Tuesday sold 15 more than Monday. Wednesday sold twice as many as Monday. Thursday sold 10 fewer than Wednesday. They sold 185 pencils in all.**

(a) Monday: ☐ (b) Tuesday: ☐ (c) Wednesday: ☐

(d) Thursday: ☐

Draw your bars:

_____

Hint. Draw Monday as 1 bar. Then draw the other days with it.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Count the squares of every size in a 4 by 4 grid. Begin with the small ones.**

(a) 1 by 1 squares: ☐ (b) 2 by 2 squares: ☐

(c) 3 by 3 squares: ☐ (d) All the squares: ☐

Hint. A 2 by 2 square can sit in 3 places across and 3 down.

2 **Beyond Level 3: A balance puzzle.**
9 coins look the same. One is lighter. You have a balance scale.

(a) With 3 coins, you can always find the light coin in ☐ weighing.

(b) With 9 coins, you can always find it in ☐ weighings.

Hint. Put 3 coins on each side. What does the scale tell you?

3 **Ali has 8 coins. They are 10¢ coins and 20¢ coins, worth $1.50 in all. Find how many coins of each kind.**

(a) ☐ ten-cent coins

(b) ☐ twenty-cent coins

10¢ coins	20¢ coins	value

Hint. Try 4 of each. How much are they worth?

4 **Make up a logic puzzle with 3 friends and 3 sports. Write 3 clues. Only ONE answer may fit. Swap with a grown-up.**

My clues: _____

_____

_____

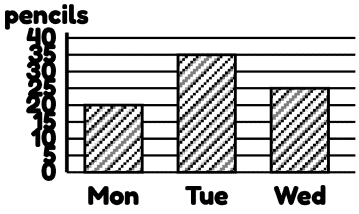
Hint. Pick the answer first. Then write clues that rule out the others.

✔ Check yourself

No hints. Try each one on your own.

1 The graph shows the pencils sold on 3 days.

Tuesday sold ☐ more pencils than Monday.



2 In a line, 3 children stand in front of Ben and 4 children stand behind him.

The line has ☐ children.

Working: _____

3 Use the cards 1, 4 and 7 once each to make 3-digit numbers. ☐ of the numbers are even.

Working: _____

4 Look for a pattern. 2, 6, 10, 14, … The 8th number is ☐.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

_____

You finished the Math Inquirers Extension Workbook, Level 3.

_____ _____

Date Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Number sleuths

Your turn
page 6

Answer: 8314
Why: An even number ends in 8 or 4. With 4 last, the other cards in order give 8314. With 8 last, the best is 4318. So 8314 is greater.

START HERE

1
page 7

Answer: (a) 4072 (b) 70
Why: Read the places from the left: 4 thousands, 0 hundreds, 7 tens and 2 ones make 4072. The 7 is in the tens place, so it is worth 70.

2
page 7

Answer: (a) 6509 (b) 2150 (c) 6203
Why: (a) 6000 + 500 + 9 = 6509. (b) 15 tens is 150, so 2000 + 150 = 2150. (c) 12 hundreds is 1200, so 5000 + 1200 + 3 = 6203.

3
page 7

Answer: (a) 7457 (b) 2950 (c) 9005 (d) 6999
Why: (a) Only the thousands digit changes: 7457. (b) 3050 - 100 = 2950. (c) 8995 + 10 = 9005, and the 9 hundreds carry. (d) 7000 - 1 = 6999.

4
page 7

Answer: 5038, 5083, 5380, 5830
Why: All four start with 5 thousands. Compare the hundreds next: 0, 0, 3, 8. Then compare the tens of 5038 and 5083: 3 is less than 8.

5
page 7

Answer: 4406
Why: 3 boxes are 3000 pins. 14 packs are 1400 pins. 3000 + 1400 + 6 = 4406.

6
page 8

Answer: (a) 8752 (b) 2578 (c) 8725 (d) 2587
Why: (a) Greatest digits first: 8752. (b) Smallest digits first, no 0 here: 2578. (c) The ones digit must be 7 or 5. With 5 last, 8725 beats 8527. (d) With 7 last, 2587 beats 2785.

7
page 8

Answer: 3005, 3050, 3500, 5003, 5030, 5300
Why: The first digit is 3 or 5, never 0. After 3, the 5 can go in the hundreds, tens or ones place: 3005, 3050, 3500. After 5, the same way: 5003, 5030, 5300. That is 6 numbers.

8
page 8

Answer: (a) 3200, 3500 (add 300) (b) 8250, 8000 (subtract 250) (c) 18, 24
Why: (a) Add 300 each time. (b) Subtract 250 each time. (c) The jumps are 1, 2, 3, 4, so the next jumps are 5 and 6: 13 + 5 = 18 and 18 + 6 = 24.

9
page 8

Answer: (a) No (b) 7000
Why: The pattern goes 4000, 4500, 5000, 5500, 6000, 6500, 7000. It jumps over 6800 and lands on 7000.

10
page 9

Answer: (a) 6150 (b) 6450 (c) 6350
Why: Each small mark is 50. A is halfway from 6100 to 6200: 6150. B is halfway from 6400 to 6500: 6450. C is halfway from 6300 to 6400: 6350.

11 page 9

Answer: 6309
Why: The thousands digit is 6. The hundreds digit is 6 - 3 = 3. The tens digit is 0. The greatest odd digit is 9. The number is 6309.

12 page 9

Answer: (a) No **(b)** 6429
Why: 9642 ends in 2, so it is even. The only odd card is 9, so 9 must be in the ones place. The other cards in order are 6, 4, 2. The number is 6429.

13 page 9

Answer: (a) 700 **(b)** 7 **(c)** 693
Why: The first 7 is in the hundreds place: 700. The second 7 is in the ones place: 7. 700 - 7 = 693.

THINK HARDER

1 page 10

Answer: (a) 5210 **(b)** 1034
Why: Only two sets of 4 different digits add up to 8: 0, 1, 2, 5 and 0, 1, 3, 4. Even numbers need an even last digit. The greatest even number is 5210. The smallest is 1034.

2 page 10

Answer: 40
Why: From 1 to 99: the digit 7 is in the ones place 10 times (7, 17, ... 97) and in the tens place 10 times (70 to 79). That is 20. From 100 to 200 it is 20 again. Total 40.

3 page 10

Answer: 2112
Why: The list goes 1111, 1112, 1121, 1122, 1211, 1212, 1221, 1222. That is 8 numbers, all starting with 1. Next are 2111 (9th) and 2112 (10th).

GO FURTHER

1 page 11

Answer: (a) 10 **(b)** 15
Why: Sum 4: first digit 4 gives 1 number (400), 3 gives 2 (310, 301), 2 gives 3, 1 gives 4. That is 1 + 2 + 3 + 4 = 10. Sum 5 goes up to 5 numbers for first digit 1: 1 + 2 + 3 + 4 + 5 = 15.

2 page 11

Answer: 50, 68
Why: The jumps are 3, 6, 9, 12. Each jump is 3 more than the one before. The next jumps are 15 and 18: 35 + 15 = 50 and 50 + 18 = 68.

3 page 11

Answer: Answers vary. Example: **I am between 7000 and 8000. My hundreds digit is double my tens digit. My ones digit is 1 more than my tens digit. My digits add up to 16. The only number is 7423.**
Why: Check that every clue is true for your secret number. Then list the numbers that fit clue 1, and cross out those that fail clue 2, and so on. In the example, the tens digit t gives 7 + 2t + t + (t + 1) = 16, so t = 2 and the number is 7423.

4 page 11

Answer: 3804
Why: The thousands digit is 3 and the tens digit is 0. If the ones digit is o, the hundreds digit is 2o, so 3 + 2o + 0 + o = 15 and o = 4. The hundreds digit is 8. The number is 3804.

CHECK YOURSELF

1 page 12
Answer: 200
Why: The 2 is in the hundreds place, so it is worth 200.

2 page 12
Answer: 5000, 4500
Why: Subtract 500 each time: 5500 - 500 = 5000 and 5000 - 500 = 4500.

3 page 12
Answer: 1056
Why: The first digit cannot be 0, so use 1. Then the rest in order: 0, 5, 6. The number is 1056.

4 page 12
Answer: 6208
Why: 5 boxes are 5000. 12 bags are 1200. 5000 + 1200 + 8 = 6208.

Unit 2 · Cryptarithm crackers

Your turn page 13
Answer: 4
Why: Ones: 8 + 7 = 15, write 5 and carry 1. Tens: 4 + 1 + 1 = 6. Hundreds: ? + 5 makes 9, so ? is 4. Thousands: 3 + 2 = 5.

START HERE

1 page 14
Answer: (a) 3 (b) 6
Why: (a) Ones 6 + 1 = 7 and tens 4 + 5 = 9 are fine. Hundreds: ? + 2 = 5, so ? is 3. (b) Ones: 9 + 5 = 14, carry 1. Tens: ? + 3 + 1 ends in 0, so ? is 6 and 10 is made. Check: 2669 + 1435 = 4104.

2 page 14
Answer: (a) 7 (b) 1
Why: Turn each one round and add. (a) 3687 + 2718 = 6405, so the digit is 7. (b) 4513 + 4617 = 9130, so the digit is 1.

3 page 14
Answer: (a) 6843 (b) 907
Why: (a) 3875 + 2968 = 6843. (b) 3875 - 2968 = 907. Check: 2968 + 907 = 3875.

4 page 14
Answer: 2375
Why: First add what he used: 2475 + 3150 = 5625. Then 8000 - 5625 = 2375. Check: 5625 + 2375 = 8000.

5 page 14
Answer: (a) 5110 (b) No (c) 3232
Why: The check should give back 5000, but 3342 + 1768 = 5110. So Siti is not right. 5000 - 1768 = 3232. Check: 3232 + 1768 = 5000.

6 page 14
Answer: (a) 7358 (b) 7359
Why: (a) Each digit and its partner add up to 9: 2 and 7, 6 and 3, 4 and 5, 1 and 8. That gives 7358. (b) 10 000 is 1 more than 9999, so the answer is 1 more than 7358.

7 page 15

Answer: (a) 1050 (b) A = 400, B = 250, C = 500
Why: The middle row is complete: 550 + 350 + 150 = 1050. Top row: 200 + 450 + A = 1050, so A = 400. Left column: 200 + 550 + 300 = 1050. Middle column: 450 + 350 + B = 1050, so B = 250. Bottom row: 300 + 250 + C = 1050, so C = 500.

8 page 15

Answer: (a) 750 (b) 480
Why: P and 830 make the top brick: 830 + P = 1580, so P = 750. Q and 270 make P: Q + 270 = 750, so Q = 480. Check: 350 + 480 = 830.

9 page 15

Answer: (a) 6 (b) 9
Why: Ones: 2 + 6 = 8. Tens: 5 + ? ends in 4, so ? is 9 and 1 is carried. Hundreds: ? + 4 + 1 ends in 1, so ? is 6 and 1 is carried. Thousands: 3 + 2 + 1 = 6. Check: 3652 + 2496 = 6148.

10 page 15

Answer: (a) 3 (b) 4
Why: (a) Try A = 3: 33 + 23 = 56. It works. (b) Two equal numbers add up to 90, so each is 45. Then C = 4.

11 page 16

Answer: (a) 5 (b) 4
Why: Turn it round: 3657 + 2845 = 6502 (the missing digit of the number taken away is its tens digit 4). Or subtract by columns: 12 - 5 = 7, then 10 - 1 - 4 = 5, then 15 - 1 - 8 = 6, then 6 - 1 - 2 = 3. The top number is 6502.

12 page 16

Answer: 5525
Why: 9000 - 3475 = 5525. Check: 3475 + 5525 = 9000.

13 page 16

Answer: (a) 1595 (b) 605
Why: For the greatest sum, put the 2 greatest digits in the hundreds places. Then the next 2 in the tens places, and the last 2 in the ones places: 800 + 700 + 50 + 40 + 3 + 2 = 1595. For the smallest sum, do the same with the smallest digits first: 200 + 300 + 40 + 50 + 7 + 8 = 605.

14 page 16

Answer: (a) 6095 (b) 640
Why: (a) 1875 + 2430 = 4305, and 4305 + 1790 = 6095. (b) 2430 - 1790 = 640. Check: 1790 + 640 = 2430.

THINK HARDER

1 page 17

Answer: (a) 5 (b) 1 (c) 9 (d) 0
Why: Two 2-digit numbers make less than 200, so B = 1 and the answer starts 11. Ones: B + C = 1 + C ends in D. Tens: A + A + carry makes 11, so A = 5 with carry 1. Then 1 + C must be 10, so C = 9 and D = 0. Check: 51 + 59 = 110.

2 page 17

Answer: 194
Why: The smallest digit, 1, goes in the hundreds place. The next smallest, 3 and 5, go in the two tens places. The others, 6 and 8, go in the ones places. For example 136 + 58 = 194, and every way of placing the digits like this gives 194.

3
page 17

Answer: (a) 630 (b) 240 (c) 300
Why: Add the middle row, the middle column and the 2 diagonals. The middle number is counted 4 times, and every other number once, so 4 totals = all 9 numbers + 3 times the middle number. All 9 numbers make 3 totals, so the total is 3 times the middle: 3 x 210 = 630. Then A = 630 - 120 - 270 = 240 and E = 630 - 120 - 210 = 300.

GO FURTHER

1
page 18

Answer: (a) 2 (b) 6 (c) 3
Why: Ones: C + C ends in B. Tens: B + B (+ carry) ends in A. Hundreds: A (+ carry) is C. Try C = 3: then B = 6, carry 0. Tens: 6 + 6 = 12, so A = 2 and carry 1. Hundreds: 2 + 1 = 3 = C. Check: 263 + 63 = 326. No other digits work.

2
page 18

Answer: 440
Why: In this pyramid the end bricks of the bottom row count once in the top brick and the 2 middle bricks count 3 times each. So 200 + 3 X + 3 X + 440 = 3280. That is 640 + 6 X = 3280, so 6 X = 2640 and X = 440. Check: the next row is 640, 880, 880; then 1520, 1760; then 3280.

3
page 18

Answer: (a) 99 (b) 143 (c) 121 (d) 11
Why: 99 = 9 x 11, 143 = 13 x 11, 121 = 11 x 11. Here is why. A 2-digit number with digits A and B is 10 A + B. Its reverse is 10 B + A. Together they make 11 A + 11 B, which is 11 x (A + B). So the answer is always in the 11 times table.

4
page 18

Answer: Answers vary. Example: top brick 800, middle left brick 370, bottom right brick 180 shown. The hidden bricks are 430 (middle right), 250 (bottom middle) and 120 (bottom left).
Why: Check that your pyramid adds up correctly and that the 3 shown bricks fix all the others. In the example: 800 - 370 = 430, then 430 - 180 = 250, then 370 - 250 = 120.

CHECK YOURSELF

1
page 19

Answer: 5
Why: Ones: 2 + 9 = 11, carry 1. Tens: 7 + 1 + 1 = 9. Hundreds: ? + 5 ends in 0, so ? is 5 and 1 is carried. Thousands: 4 + 2 + 1 = 7.

2
page 19

Answer: 2
Why: Turn it round: 4635 + 3569 = 8204, so the missing digit is 2.

3
page 19

Answer: 520
Why: M is the sum of 240 and 280: 240 + 280 = 520. Check: 520 + 640 = 1160.

4
page 19

Answer: 2895
Why: 2845 + 3260 = 6105. Then 9000 - 6105 = 2895. Check: 6105 + 2895 = 9000.

Unit 3 · Times table treasure

Your turn
page 20

Answer: 48
Why: 3 × 8 = 24 and 3 × 8 = 24. Then 24 + 24 = 48. So 6 × 8 = 48.

START HERE

1
page 21

Answer: 32, 48, 72
Why: The answers go up by 8 each time. 8 × 4 = 32, 8 × 6 = 48 and 8 × 9 = 72.

2
page 21

Answer: (a) 48 (b) 63 (c) 54 (d) 56
Why: 6 × 8 = 48, 7 × 9 = 63, 9 × 6 = 54 and 8 × 7 = 56. Swap the numbers to check: 8 × 6 = 48.

3
page 21

Answer: (a) 48 (b) 32 (c) 80 (d) 72
Why: (a) One more 8: 40 + 8 = 48. (b) One less 8: 40 - 8 = 32. (c) Double 8 × 5: 80. (d) 8 × 10 = 80, take away one 8: 72.

4
page 21

Answer: (a) 63 (b) 8
Why: (a) 9 × 7 = 63. (b) 56 ÷ 7 = 8, because 7 × 8 = 56.

5
page 21

Answer: 63
Why: The numbers in the 7 times table between 50 and 80 are 56, 63, 70 and 77. Their digits add up to 11, 9, 7 and 14. Only 63 has digits that add up to 9.

6
page 22

Answer: (a) 45 (b) 18 (c) 63
Why: 5 × 9 = 45 and 2 × 9 = 18. Add: 45 + 18 = 63. So 7 × 9 = 63.

7
page 22

Answer: A = 48, B = 42, C = 63, D = 56
Why: Each cell is the row number × the column number. A = 6 × 8 = 48. B = 7 × 6 = 42. C = 7 × 9 = 63. D = 8 × 7 = 56.

8
page 22

Answer: Columns 6 and 7. Rows 9 and 8.
Why: 54 and 63 are in the same row. Both are in the 9 times table: 54 = 9 × 6 and 63 = 9 × 7. So the row number is 9 and the columns are 6 and 7. The second row: 48 = 8 × 6 and 56 = 8 × 7, so the row number is 8.

9
page 23

Answer: (a) 8 (b) 6 (c) 5 (d) 6
Why: Only the last digits matter. (a) 6 × 3 = 18 ends in 8. (b) 8 × 7 = 56 ends in 6. (c) 9 × 5 = 45 ends in 5. (d) 7 × 8 = 56 ends in 6.

10
page 23

Answer: 6
Why: Check the number of rows from 2 upwards: 2 rows of 15, 3 rows of 10, 5 rows of 6, 6 rows of 5, 10 rows of 3 and 15 rows of 2. 4 rows does not work because 30 is not in the 4 times table. That is 6 ways.

11
page 23

Answer: 24, 42, 60
Why: 6 × 4 = 24, 6 × 7 = 42 and 6 × 10 = 60. The numbers 38 and 56 are not: 6 × 6 = 36 and 6 × 9 = 54, and the next answer after 54 is 60.

12 page 23
Answer: (a) Ben (b) 1
Why: Mei has 6 × 8 = 48 stickers. Ben has 7 × 7 = 49 stickers. Ben has 49 - 48 = 1 more.

THINK HARDER

1 page 24
Answer: 3
Why: 7 × 2 = 14 ends in 4. The last digits of the 7 times table repeat every 10 numbers, so the next ones are 7 × 12 = 84 and 7 × 22 = 154. 7 × 32 is beyond 7 × 30. That makes 3 answers.

2 page 24
Answer: (a) 54 (b) 56 (c) 56 (d) 64
Why: 49 = 7 × 7, so row 2 and column 1 are both 7. Then 42 = 6 × 7 gives row 1 = 6, and 48 = 6 × 8 gives column 2 = 8. 63 = 7 × 9 gives column 3 = 9. 72 = 8 × 9 gives row 3 = 8. A = 6 × 9 = 54, B = 7 × 8 = 56, C = 8 × 7 = 56 and D = 8 × 8 = 64.

3 page 24
Answer: 7 and 9
Why: The pairs that multiply to 63 are 1 and 63, 3 and 21, 7 and 9. Their sums are 64, 24 and 16. Only 7 and 9 add up to 16.

GO FURTHER

1 page 25
Answer: 738, 963, 1089
Why: 738: 7 + 3 + 8 = 18, then 1 + 8 = 9. 825: 8 + 2 + 5 = 15, then 1 + 5 = 6, so not. 963: 9 + 6 + 3 = 18, then 9. 1089: 1 + 0 + 8 + 9 = 18, then 9. Check: 738 = 9 × 82, 963 = 9 × 107, 1089 = 9 × 121.

2 page 25
Answer: (a) 21 (b) 112
Why: The seats go 7, 9, 11, 13, 15, 17, 19, 21. Row 8 has 21 seats. Pair them: 7 + 21 = 28, 9 + 19 = 28, 11 + 17 = 28, 13 + 15 = 28. There are 4 pairs, so 4 × 28 = 112 seats.

3 page 25
Answer: Answers vary. Example: the products 42, 48 and 49 are shown. The row numbers are 6 and 7 and the column numbers are 7 and 8, so the hidden product is 56.
Why: Check that each shown product belongs to the numbers you chose. Check that no other row and column numbers from 6 to 9 give the same 3 products. In the example, 42 = 6 × 7 and 48 = 6 × 8 give row 6 and columns 7 and 8. Also 49 = 7 × 7 gives row 7. The hidden product is 7 × 8 = 56.

4 page 25
Answer: (a) 2 or 7 (b) none
Why: The last digits of the 8 times table are 8, 6, 4, 2, 0, 8, 6, 4, 2, 0. A last digit of 6 comes from 8 × 2 and 8 × 7. A last digit of 5 never comes: every answer in the 8 times table is even.

CHECK YOURSELF

1 page 26
Answer: 56
Why: 7 × 8 = 56. Check: 8 × 7 = 56.

2 page 26
Answer: 42
Why: 6 × 7 is one more 7 than 5 × 7: 35 + 7 = 42.

3 page 26
Answer: 2
Why: Only 7 × 6 matters for the last digit. 7 × 6 = 42 ends in 2. (7 × 56 = 392.)

4 page 26
Answer: 9
Why: 54 ÷ 6 = 9, because 6 × 9 = 54.

Unit 4 · Model makers

Your turn page 27
Answer: 265
Why: Ben has fewer, so subtract: 480 - 215 = 265. Check: 265 + 215 = 480.

START HERE

1 page 28
Answer: 1525
Why: The whole bar is 2400 eggs. One part is 875, so the white part is 2400 - 875 = 1525. Check: 875 + 1525 = 2400.

2 page 28
Answer: (a) 1070 (b) 2520
Why: (a) Aisha has fewer, so subtract: 1450 - 380 = 1070. (b) Together: 1450 + 1070 = 2520.

3 page 28
Answer: (a) 1399 (b) 147
Why: (a) 468 + 539 = 1007, and 1007 + 392 = 1399. (b) 539 - 392 = 147.

4 page 28
Answer: (a) 35 (b) 35
Why: Before: 64 - 29 = 35. After: Siti has 76 and Wei has 41, and 76 - 41 = 35. The gap stays the same because both children get the same number of beads.

5 page 29
Answer: (a) 35 (b) 105
Why: There are 4 equal boxes in all. 140 ÷ 4 = 35, so 1 box is 35. Aisha has 1 box: 35 coins. Ravi has 3 boxes: 3 × 35 = 105 coins.

6 page 29
Answer: (a) 36 (b) 54
Why: Take away the extra 18 from the total: 90 - 18 = 72. Share the 72 into 2 equal parts: 36 each. Wei has 36. Kumar has 36 + 18 = 54. Check: 36 + 54 = 90.

7 page 29
Answer: 13
Why: The gap is 78 - 52 = 26 cm. The piece closes the gap from both sides, so it is half of the gap: 26 ÷ 2 = 13 cm. Check: 78 - 13 = 65 and 52 + 13 = 65.

8 page 30
Answer: (a) 113 (b) 96 (c) 294
Why: (a) 85 + 28 = 113. (b) 113 - 17 = 96. (c) 85 + 113 + 96 = 294. Wednesday is compared with Tuesday, not with Monday.

9 page 30
Answer: 24.85
Why: First add what she spends: 18.40 + 6.75 = 25.15. Then 50.00 - 25.15 = 24.85. Check: 25.15 + 24.85 = 50.00.

10 page 30
Answer: (a) 16 (b) 64
Why: Wei has 3 boxes and Aisha has 1 box. 3 boxes are 48 marbles, so 1 box is 48 ÷ 3 = 16. Together they have 4 boxes: 4 × 16 = 64.

11 page 30
Answer: (a) No (b) 36
Why: Ben has fewer stickers than Ali, so Ben has less than 54. Ravi added instead of subtracting. The right answer is 54 - 18 = 36.

12 page 30
Answer: 49
Why: In 7 weeks she saves 7 × $12 = $84. Then 84 - 35 = 49. She has $49 left.

THINK HARDER

1 page 31
Answer: 21
Why: At first there are 4 equal boxes: 84 ÷ 4 = 21, so Wei has 21 and Kumar has 63. At the end each has 84 ÷ 2 = 42. Kumar gives 63 - 42 = 21 marbles. Check: 21 + 21 = 42.

2 page 31
Answer: (a) 3 (b) 8
Why: The second bill has 3 more pens and costs $23 - $14 = $9 more. So 3 pens cost $9 and 1 pen costs $3. The box costs $14 - 2 × $3 = $8. Check: 8 + 15 = 23.

3 page 31
Answer: 21
Why: The gap is 35 - 7 = 28 years and it never changes. When the father is twice as old, Ravi is 1 box and his father is 2 boxes, so 1 box = the gap = 28. Ravi is 28, which is 21 years from now. Check: 28 and 56.

GO FURTHER

1 page 32
Answer: (a) 26 (b) 38 (c) 76
Why: Lily is 1 box. Kumar is 1 box and 12. Aisha is 2 boxes and 24. In all: 4 boxes and 36 = 140. So 4 boxes = 104 and 1 box = 26. Lily 26, Kumar 38, Aisha 76. Check: 26 + 38 + 76 = 140.

2 page 32
Answer: 30
Why: Red is 4 boxes and blue is 1 box. After selling 18 red balloons, red is 1 box. So 3 boxes are 18, and 1 box is 6. At first there were 6 blue and 24 red: 30 balloons in all.

3 page 32
Answer: Answers vary. Example: Ali and Ben have 60 stickers in all. Ben has 12 more stickers than Ali. Ali has 24 stickers and Ben has 36 stickers.
Why: Check that the story says 60 in all and 12 more for Ben, and that the answer works. In the example, take away the 12: 60 - 12 = 48, and 48 ÷ 2 = 24. Ali has 24 and Ben has 24 + 12 = 36.

CHECK YOURSELF

1 page 33
Answer: 625
Why: 2000 - 735 = 1265. Then 1265 - 640 = 625. Check: 735 + 640 + 625 = 2000.

2 page 33
Answer: 1255
Why: Siti has 560 + 135 = 695 beads. Together: 560 + 695 = 1255.

3 page 33
Answer: 60
Why: There are 6 equal boxes in all. 72 ÷ 6 = 12, so 1 box is 12. Siti has 5 boxes: 5 × 12 = 60.

4 page 33
Answer: 29
Why: Take away the extra 8: 50 - 8 = 42. Share 42 into 2 equal parts: 21 each. Box B has 21 and Box A has 21 + 8 = 29.

Unit 5 · Remainder riddles

Your turn page 34
Answer: 7 full bracelets, 1 bead left over
Why: 6 × 7 = 42 and 43 - 42 = 1. So 43 ÷ 6 = 7 remainder 1. Check: 7 × 6 + 1 = 43.

START HERE

1 page 35
Answer: (a) 4 (b) 3
Why: Count the solid boxes: 4 groups of 4 dots make 16 dots. The dashed box has the 3 dots left over. So 19 ÷ 4 = 4 remainder 3.

2 page 35
Answer: (a) 5 remainder 3 (b) 6 remainder 2 (c) 6 remainder 7 (d) 12 remainder 3
Why: (a) 7 × 5 = 35 and 38 - 35 = 3. (b) 8 × 6 = 48 and 50 - 48 = 2. (c) 9 × 6 = 54 and 61 - 54 = 7. (d) 6 × 12 = 72 and 75 - 72 = 3.

3 page 35
Answer: (a) 6 (b) 6 (c) 7 (d) 2
Why: (a) 26 ÷ 5 = 5 remainder 1. The 1 person left over still needs a car, so 6 cars. (b) 37 ÷ 6 = 6 remainder 1: 6 whole pieces. (c) 30 ÷ 4 = 7 remainder 2: 7 toys. (d) The remainder is the money left: $2.

4 page 35
Answer: (a) 7 (b) 8 remainder 3
Why: A remainder must be smaller than the number we divide by. 10 is greater than 7, so another group of 7 fits. 7 × 8 = 56 and 59 - 56 = 3. So 59 ÷ 7 = 8 remainder 3.

5 page 36
Answer: (a) 58 (b) 8 bags, 2 marbles left
Why: (a) 9 × 6 = 54, and 54 + 4 = 58. (b) 58 ÷ 7: 7 × 8 = 56 and 58 - 56 = 2. So 8 full bags and 2 marbles left.

6 page 36
Answer: (a) 0, 1, 2, 3, 4 (b) 71
Why: (a) A remainder is always smaller than 5, so it can be 0, 1, 2, 3 or 4. (b) The greatest remainder for 9 is 8. The numbers with remainder 8 are 8, 17, 26, ... 62, 71, 80. Only 71 is between 70 and 80.

7 page 36
Answer: (a) 0, 1, 2, 3, 0, 1, 2, 3 (b) 2 (c) 0
Why: The remainders go 0, 1, 2, 3 and then start again. (b) 30 = 28 + 2, and 28 is in the 4 times table, so the remainder is 2. (c) 100 = 4 × 25, so the remainder is 0.

8 page 36

Answer: (a) 31 and 37 (b) 31
Why: Between 30 and 40, 6 × 5 = 30 and 6 × 6 = 36, so the numbers are 30 + 1 = 31 and 36 + 1 = 37. Now test them with 4: 31 ÷ 4 = 7 remainder 3 (right). 37 ÷ 4 = 9 remainder 1 (not right). The number is 31.

9 page 37

Answer: (a) 6 (b) 7 (c) 2
Why: 52 ÷ 9 = 5 remainder 7. 5 full buses are not enough, because 7 children are left. So they need 6 buses. The last bus has 7 children, and 9 - 7 = 2 seats are empty.

10 page 37

Answer: (a) B (b) D
Why: The pattern repeats every 4 letters. 30 ÷ 4 = 7 remainder 2, so the 30th letter is the 2nd letter of the pattern: B. 32 ÷ 4 = 8 with no remainder, so the 32nd letter is the last letter of the pattern: D.

11 page 37

Answer: (a) 77 (b) 8 beads each, 5 left over
Why: (a) 8 × 9 = 72, and 72 + 5 = 77. (b) 77 ÷ 9: 9 × 8 = 72 and 77 - 72 = 5. Each friend gets 8 beads and 5 beads are left over.

12 page 37

Answer: 18, 23, 33 and 48
Why: A number leaves remainder 3 when it is 3 more than a number in the 5 times table. 18 = 15 + 3, 23 = 20 + 3, 33 = 30 + 3 and 48 = 45 + 3. 41 is 1 more than 40, so it leaves remainder 1.

13 page 37

Answer: (a) 6 (b) 3
Why: (a) 53 ÷ 8 = 6 remainder 5, so she makes 6 full cards. (b) 5 stickers are left over, so she needs 8 - 5 = 3 more to fill another card.

THINK HARDER

1 page 38

Answer: (a) Fri (b) Wed
Why: (a) 45 ÷ 7 = 6 remainder 3. After 6 weeks it is Tuesday again. Then 3 more days: Wednesday, Thursday, Friday. (b) 20 ÷ 7 = 2 remainder 6. Go back 2 weeks to Tuesday, then back 6 more days: Wednesday.

2 page 38

Answer: 67
Why: The 2-digit numbers that leave remainder 2 when divided by 5 end in 2 or 7. Test them with 3 and 4. 67 = 66 + 1 (66 is in the 3 times table) and 67 = 64 + 3 (64 is in the 4 times table). No other number works.

3 page 38

Answer: (a) 9 (b) 38
Why: Giving each cousin 2 more crayons uses the 11 crayons that were not used and needs 7 more: 11 + 7 = 18 crayons. So 18 ÷ 2 = 9 cousins. The crayons: 9 × 3 + 11 = 38. Check: 9 × 5 = 45 and 45 - 38 = 7 more are needed.

GO FURTHER

1 page 39

Answer: (a) 1 (b) 6 (c) 2
Why: (a) 3 + 5 = 8, and 8 ÷ 7 leaves remainder 1. Test: 10 + 5 = 15 and 15 ÷ 7 = 2 remainder 1. (b) 3 + 3 = 6, which is smaller than 7. Test: 20 ÷ 7 = 2 remainder 6. (c) 5 + 5 = 10, and 10 ÷ 8 leaves remainder 2. Test: 13 doubled is 26, and 26 ÷ 8 = 3 remainder 2.

2 page 39
Answer: 58
Why: 1 left over when counting in 3s is the same as being 2 short of a full group. Counting in 4s with 2 left is 2 short too, and counting in 5s with 3 left is 2 short. So the number is 2 less than a number in the 3, 4 and 5 times tables. The first such number is 60 (and then 120). 60 - 2 = 58. Check: 58 ÷ 3 = 19 r 1, 58 ÷ 4 = 14 r 2, 58 ÷ 5 = 11 r 3.

3 page 39
Answer: Answers vary. Example: I am between 10 and 25. When I am divided by 4 the remainder is 1. When I am divided by 3 the remainder is 2. The only number is 17.
Why: Check that your number gives the remainders you wrote, and list the numbers that fit your first clue. Test each with the second clue. In the example, the numbers with remainder 1 when divided by 4 are 13, 17 and 21. Only 17 leaves remainder 2 when divided by 3.

4 page 39
Answer: (a) Yes (b) 1, 2, 4 and 7
Why: Orders he can fill: 3, 5, 6 (3 + 3), 8 (3 + 5), 9, 10, 11 (3 + 3 + 5), 12, 13 (3 + 5 + 5), 14 (3 + 3 + 3 + 5), 15. So the orders he cannot fill are 1, 2, 4 and 7.

CHECK YOURSELF

1 page 40
Answer: 6 remainder 4
Why: 8 × 6 = 48 and 52 - 48 = 4. So 52 ÷ 8 = 6 remainder 4.

2 page 40
Answer: 5
Why: 40 ÷ 9 = 4 remainder 4. The 4 people left over need another bus: 5 buses.

3 page 40
Answer: 1
Why: 6 × 4 = 24 and 25 - 24 = 1. One muffin is left over.

4 page 40
Answer: 29
Why: The numbers from 20 to 30 that leave remainder 2 when divided by 3 are 20, 23, 26 and 29. Of these, only 29 leaves remainder 4 when divided by 5 (25 + 4).

Unit 6 · Fraction walls

Your turn page 41
Answer: <
Why: 2/3 = 8/12 on the wall. 7/12 is less than 8/12, so 7/12 < 2/3.

START HERE

1 page 42
Answer: (a) 1/3 = 2/6 = 4/12 (b) 2/3 = 4/6 = 8/12 (c) 5/6 = 10/12
Why: Line up the rows. 1 third is the same length as 2 sixths and 4 twelfths. 2 thirds is 4 sixths and 8 twelfths. 5 sixths is 10 twelfths.

2 page 42
Answer: (a) 3/8 (b) 4/6 (c) 7/10 (d) 2/5
Why: The bottom number is the number of equal parts in the whole shape. The top number is the number of shaded parts. A: 3 of 8. B: 4 of 6. C: 7 of 10. D: 2 of 5.

3 page 42
Answer: A and C
Why: A fraction needs equal parts. A has 4 equal parts and C has 6 equal sectors. B and D have parts of different sizes.

4 page 43
Answer: (a) > (b) > (c) = (d) <
Why: (a) Same bottom number: 5 parts are more than 3 parts. (b) Thirds are larger pieces than quarters. (c) 2/4 = 4/8. (d) 3/4 = 9/12 and 5/6 = 10/12, and 9 is less than 10.

5 page 43
Answer: 1/3, 1/2, 2/3, 5/6
Why: Write each one in sixths: 5/6, 2/6, 3/6, 4/6. The order of the tops is 2, 3, 4, 5. So the order is 1/3, 1/2, 2/3, 5/6.

6 page 43
Answer: (a) 7 (b) 3 (c) 5 (d) 7
Why: When the bottom numbers are the same, add or subtract the top numbers and keep the bottom number. (d) 1 whole is 10/10, and 10/10 - 3/10 = 7/10.

7 page 43
Answer: (a) 3 (b) 3 (c) 5 (d) 2
Why: (a) 1/2 = 2/4, so 2/4 + 1/4 = 3/4. (b) 1/3 = 2/6, so 2/6 + 1/6 = 3/6. (c) 3/4 = 6/8, so 6/8 - 1/8 = 5/8. (d) 1/2 = 3/6, so 5/6 - 3/6 = 2/6.

8 page 44
Answer: (a) 7 (b) 5 (c) 1
Why: (a) 3/12 + 4/12 = 7/12. (b) The whole cake is 12/12, so 12/12 - 7/12 = 5/12. (c) 4/12 - 3/12 = 1/12.

9 page 44
Answer: (a) 3 (b) 5 (c) 7 (d) D
Why: There are 8 equal steps, so each step is 1/8. A is 3 steps from 0: 3/8. B is 5 steps: 5/8. C is 7 steps: 7/8. D is 4 steps: 4/8, which is half of the line.

10 page 44
Answer: (a) 9 (b) 4 (c) 10 (d) 4
Why: Multiply or divide the top and the bottom by the same number. (a) 4 × 3 = 12, so 3 × 3 = 9. (b) 5 × 2 = 10, so 2 × 2 = 4. (c) 1 × 5 = 5, so 2 × 5 = 10. (d) 6 ÷ 2 = 3, so 8 ÷ 2 = 4.

11 page 44
Answer: (a) No (b) 5
Why: When the parts are the same size, only the top numbers are added. The bottom number stays 8. 3/8 + 2/8 = 5/8.

THINK HARDER

1 page 45
Answer: 5/12, 6/12 and 7/12
Why: 1/3 = 4/12 and 2/3 = 8/12. The twelfths between 4/12 and 8/12 are 5/12, 6/12 and 7/12. The fractions 4/12 and 8/12 are not between.

2 page 45
Answer: 6/15
Why: Equal fractions of 2/5 are 4/10, 6/15, 8/20. The gap between bottom and top is 3, 6, 9, 12. The gap is 9 for 6/15. Check: 15 - 6 = 9.

3 page 45
Answer: (a) 7 (b) 5
Why: 1/4 = 3/12 and 1/3 = 4/12. (a) 3/12 + 4/12 = 7/12. (b) 12/12 - 7/12 = 5/12 of the jug is left.

GO FURTHER

1 page 46
Answer: (a) 4 (b) 3 (c) 4
Why: (a) 3/4 - 1/2 = 3/4 - 2/4 = 1/4. (b) 5/6 - 1/2 = 5/6 - 3/6 = 2/6 = 1/3. (c) 7/12 - 1/3 = 7/12 - 4/12 = 3/12 = 1/4.

2 page 46
Answer: (a) 0 halves, 2 quarters, 4 eighths
Why: Write everything in eighths: a half is 4, a quarter is 2, an eighth is 1. The 6 pieces must make 8 eighths. With no halves: 2 quarters (4) and 4 eighths (4) make 8. Using 1 half is not possible with 6 pieces: the other 5 pieces would have to make 4 eighths, but each is at least 1.

3 page 46
Answer: (a) 2 (b) 6
Why: 1/2 of the bar is 6 parts and 1/3 of the bar is 4 parts. 6 + 4 = 10 parts are coloured, so 2 parts are white. 2/12 is the same as 1/6.

4 page 46
Answer: Answers vary. Example: Ali eats 3/8 of a pizza and Mei eats 2/8 of it. Together they eat 5/8 of the pizza.
Why: Check that your fractions all have the same bottom number, and that the story gives 5/8. In the example, 3/8 + 2/8 = 5/8.

CHECK YOURSELF

1 page 47
Answer: 5
Why: The bar has 8 equal parts and 5 are shaded. The fraction is 5/8.

2 page 47
Answer: <
Why: 3/4 = 6/8. 5/8 is less than 6/8, so 5/8 < 3/4.

3 page 47
Answer: 3
Why: 1/4 = 2/8, so 5/8 - 2/8 = 3/8.

4 page 47
Answer: 8
Why: 3 × 4 = 12, so 2 × 4 = 8. 2/3 = 8/12.

Unit 7 · Area architects

Your turn page 48
Answer: Area 24 cm². Perimeter 22 cm.
Why: Area: 8 × 3 = 24 cm². Perimeter: 8 + 3 + 8 + 3 = 22 cm, or 2 × (8 + 3) = 22 cm.

START HERE

1 page 49
Answer: (a) A: 8 cm², 12 cm (b) B: 6 cm², 14 cm (c) C: 6 cm², 10 cm (d) B,C
Why: Area is the number of squares: A has 8, B has 4 + 2 = 6 and C has 6. Perimeter is the length of the edge: A is 2 × (4 + 2) = 12. B: the box round it is 4 by 2, with a perimeter of 12. The dent in the bottom adds a wall of 1 cm on each side: 12 + 2 = 14. C is 2 × (3 + 2) = 10. B and C have the same area, 6, but different perimeters.

2 page 49

Answer: (a) Area 48 cm², perimeter 28 cm (b) Area 44 m², perimeter 30 m
Why: (a) Area 8 × 6 = 48. Perimeter 2 × (8 + 6) = 28. (b) Area 11 × 4 = 44. Perimeter 2 × (11 + 4) = 30.

3 page 49

Answer: (a) 28 (b) 49 (c) 9
Why: (a) 4 × 7 = 28. (b) 7 × 7 = 49. (c) The 4 sides are equal, so each is 36 ÷ 4 = 9 cm.

4 page 49

Answer: (a) 6 (b) 54
Why: Length + width is half of the perimeter: 30 ÷ 2 = 15. So the width is 15 - 9 = 6 cm. The area is 9 × 6 = 54 cm².

5 page 50

Answer: (a) 5 (b) 3 (c) 15
Why: (a) 15 ÷ 3 = 5. (b) 9 ÷ 3 = 3. (c) 5 rows of 3 tiles make 5 × 3 = 15 tiles.

6 page 50

Answer: (a) 3 (b) 12
Why: Length + width = 14 ÷ 2 = 7. The pairs are 6 and 1 (area 6), 5 and 2 (area 10), 4 and 3 (area 12). That is 3 rectangles. The one closest to a square, 4 by 3, has the greatest area: 12 cm².

7 page 50

Answer: (a) 4 (b) 3 (c) 36 (d) 28
Why: a is the top of the lower part: 8 - 4 = 4 cm. b is the rest of the left height: 6 - 3 = 3 cm. Area: cut into a 4 by 6 rectangle and a 4 by 3 rectangle: 24 + 12 = 36 cm². Perimeter: push the steps out to a box 8 by 6: 2 × (8 + 6) = 28 cm.

8 page 51

Answer: (a) 12 (b) 18 (c) 6 (d) 18
Why: (a) 6 + 4 + 2 = 12 cm². (b) The staircase has the same perimeter as its box, 6 by 3: 2 × (6 + 3) = 18 cm. (c) The rectangle has 6 × 3 = 18 squares, so he adds 18 - 12 = 6. (d) 18 cm: filling the steps does not change the perimeter.

9 page 51

Answer: (a) 6 (b) Ben's (c) 16
Why: (a) Mei's perimeter is 2 × (10 + 2) = 24 cm, so Ben's side is 24 ÷ 4 = 6 cm. (b) Ben's area is 6 × 6 = 36 cm² and Mei's area is 10 × 2 = 20 cm². (c) 36 - 20 = 16 cm².

10 page 51

Answer: (a) A and B (b) C
Why: A square that touches the figure along only 1 side adds 2 to the perimeter. A and B each touch 1 side, so the perimeter becomes 12 cm. C fits into the corner and touches 2 sides: it hides 2 cm of the old edge and adds 2 cm, so the perimeter stays 10 cm.

11 page 51

Answer: (a) 8 (b) 32
Why: (a) 8 × 8 = 64, so each side is 8 m. (b) The border is the perimeter: 4 × 8 = 32 m.

THINK HARDER

1 page 52

Answer: (a) 24 (b) 48 (c) 16
Why: Each rectangle is 8 cm by 4 cm, so its perimeter is 2 × (8 + 4) = 24 cm. Together: 48 cm. The square's perimeter is 32 cm, and 48 - 32 = 16. The cut is 8 cm long and it appears on both pieces, so it adds 2 × 8 = 16 cm.

2 page 52

Answer: (a) 9 (b) 4
Why: Length + width is 26 ÷ 2 = 13. The pairs that multiply to 36 are 1 and 36, 2 and 18, 3 and 12, 4 and 9, 6 and 6. The pair that adds up to 13 is 4 and 9. Check: area 9 × 4 = 36 and perimeter 2 × 13 = 26.

3 page 52

Answer: (a) 48 (b) 30 (c) 34
Why: (a) 9 × 6 = 54, and 54 - 6 = 48 m². (b) Cutting a corner does not change the perimeter: 2 × (9 + 6) = 30 m. (c) A notch in the middle of a side takes away 3 m of edge. It adds 3 + 2 + 2 = 7 m of new edge. So the perimeter grows by 4 m: 34 m.

GO FURTHER

1 page 53

Answer: (a) 22 (b) 14
Why: Each new square adds 2 cm (1 cm on top and 1 cm below). The perimeter is 4 + 2 for every extra square. For 10 squares: 4 + 9 × 2 = 22 cm. For 30 cm: (30 - 4) ÷ 2 = 13 extra squares, so 14 squares.

2 page 53

Answer: (a) 8 m by 6 m (b) 24
Why: The path adds 1 m on each side, so both lengths grow by 2 m: 6 + 2 = 8 and 4 + 2 = 6. The whole rectangle has an area of 8 × 6 = 48 m². The pond has 6 × 4 = 24 m². The path is 48 - 24 = 24 m².

3 page 53

Answer: (a) 2 (b) 30
Why: The side must divide both 12 and 10 exactly. 1 and 2 do. 3 does not divide 10, 4 does not, 5 does not divide 12, and 6 does not divide 10. So the largest side is 2 m. There are 6 tiles along and 5 tiles across: 30 tiles.

CHECK YOURSELF

1 page 54

Answer: 45
Why: 9 × 5 = 45 cm².

2 page 54

Answer: 24
Why: 4 × 6 = 24 cm.

3 page 54

Answer: Area 7 cm². Perimeter 12 cm.
Why: Area: 3 + 2 + 2 = 7 cm². Perimeter: the figure fits in a 3 by 3 box and has no dents on the outside, so 2 × (3 + 3) = 12 cm.

4 page 54

Answer: 7
Why: Length + width is 22 ÷ 2 = 11. The length is 11 - 4 = 7 cm.

Unit 8 · Angle and line detectives

Your turn
page 55

Answer: smaller than
Why: The second line is inside the corner of the checker, so the angle is smaller than a right angle.

START HERE

1
page 56

Answer: (a) B and E **(b)** A and F **(c)** C and D
Why: A and F fit the checker exactly, so they are right angles. B and E are inside the corner of the checker, so they are smaller. C and D stick out past the corner of the checker, so they are greater.

2
page 56

Answer: (a) 5 **(b)** 6 **(c)** 2
Why: Shape A is an L. It has 5 right angles, and its dent is greater than a right angle. Shape B has 6 right angles and 2 dents. Shape C has right angles only at its 2 left corners. Its 2 sloping corners do not fit the checker.

3
page 56

Answer: (a) F and N **(b)** F and L
Why: F has 2 parallel arms and a stem that meets them at right angles. V has 2 sloping lines that are neither parallel nor perpendicular. N has 2 parallel upright lines, but its sloping line is not perpendicular to them. L has 2 lines that meet at a right angle.

4
page 57

Answer: (a) 3 **(b)** 2 **(c)** 1 **(d)** 6
Why: There are 3 small angles. The next size is made of 2 small angles side by side: 2 of those. Then 1 angle made of all 3 small ones. 3 + 2 + 1 = 6 angles.

5
page 57

Answer: (a) R **(b)** P
Why: Line AB goes 4 across and 2 up. R goes 4 across and 2 up too, so it never meets AB: parallel. P goes 2 across and 4 down, which turns AB through a right angle: perpendicular. Q and S slope in other ways.

6
page 57

Answer: (a) 4 **(b)** 12
Why: At each end of a rung, the rung and the rail make 2 right angles (one above the rung and one below it). A rung has 2 ends, so 4 right angles. 3 rungs make 3 × 4 = 12 right angles.

7
page 58

Answer: (a) E **(b)** W **(c)** S
Why: Going round to the right, north is followed by east, south and west. (a) 1 turn: east. (b) 3 turns: west. (c) 2 turns to the left from north: west, then south. Two right angles in either direction face the opposite way: south.

8
page 58

Answer: (a) 4 **(b)** 2 **(c)** 1 **(d)** perpendicular
Why: (a) Each corner of a rectangle is a right angle: 4. (b) The 2 long sides are parallel and the 2 short sides are parallel: 2 pairs. (c) If a triangle had 2 right angles, those sides would be parallel and never meet to make a triangle. (d) The sides of a square meet at right angles.

9 page 58
Answer: (a) True (b) False (c) True (d) False
Why: (a) That is what perpendicular means. (b) Parallel lines never meet. (c) A line across 2 rails of a ladder is perpendicular to both. (d) Lines can cross at any angle. Only some crossings make right angles.

10 page 58
Answer: (a) No (b) greater than
Why: Angle X is a dent. It points into the shape. A checker does not fit inside it: the angle is greater than a right angle.

THINK HARDER

1 page 59
Answer: (a) 10 (b) 21
Why: With 5 lines there are 4 small angles. Then 3 angles made of 2 small angles, 2 made of 3, and 1 made of 4. 4 + 3 + 2 + 1 = 10. With 7 lines the small angles are 6: 6 + 5 + 4 + 3 + 2 + 1 = 21.

2 page 59
Answer: (a) R
Why: AB goes 2 across for every 1 up. A line perpendicular to it goes 1 across and 2 down (a quarter turn). From C, 1 across and 2 down reaches the dot at 3 across and 4 up, then (4, 2), then the bottom row at R.

3 page 59
Answer: (a) 7 (b) 3 (c) 11
Why: Count the corners that fit a checker: 7. The 3 corners that dent into the shape are greater than a right angle. A staircase with n steps has n - 1 dents and n + 3 right angles. For 8 steps: 8 + 3 = 11 right angles.

GO FURTHER

1 page 60
Answer: (a) 180 (b) 360 (c) 45 (d) 270
Why: (a) 90 + 90 = 180. (b) 4 × 90 = 360. (c) 90 ÷ 2 = 45. (d) 3 × 90 = 270.

2 page 60
Answer: (a) parallel (b) Yes
Why: (a) Two lines that are perpendicular to the same line stay the same distance apart and never meet: they are parallel. (b) A line that crosses 2 parallel lines makes the same angle with both. So a right angle with one means a right angle with the other.

3 page 60
Answer: (a) 2 (b) 16 (c) 36
Why: (a) The 2 lines across are parallel and the 2 lines up are parallel: 2 pairs. (b) The lines cross in 2 × 2 = 4 places, and each crossing has 4 right angles: 16. (c) 3 × 3 = 9 crossings, 9 × 4 = 36 right angles.

CHECK YOURSELF

1 page 61
Answer: greater than
Why: The second line sticks out past the corner of the checker, so the angle is greater than a right angle.

2 page 61
Answer: E and H
Why: E has 3 parallel arms. H has 2 parallel upright lines. V has 2 sloping lines that are not parallel. T has no pair of parallel lines.

3 page 61
Answer: 2
Why: The 2 long sides are parallel and the 2 short sides are parallel.

4 page 61
Answer: N
Why: Turning left from east, the next direction is north.

Unit 9 · Measure and money masters

Your turn page 62
Answer: 1 ℓ 550 ml
Why: 2 ℓ 350 ml = 2350 ml. Then 2350 - 800 = 1550 ml = 1 ℓ 550 ml. Check: 1550 + 800 = 2350.

START HERE

1 page 63
Answer: (a) 300 (b) 4050 (c) 5300 (d) 2 kg 750 g (e) 3 m 65 cm
Why: (a) 3 × 100 = 300. (b) 4 kg = 4000 g, and 4000 + 50 = 4050. (c) 5 ℓ = 5000 ml, and 5000 + 300 = 5300. (d) 2000 g is 2 kg, and 750 g is left. (e) 300 cm is 3 m, and 65 cm is left.

2 page 63
Answer: (a) 8 (b) 7 (c) 1
Why: The pencil starts at 3 and ends at 11: 11 - 3 = 8 cm. The crayon starts at 6 and ends at 13: 13 - 6 = 7 cm. The pencil is 8 - 7 = 1 cm longer.

3 page 63
Answer: C, A, D, B
Why: Write each in grams: A is 1080 g, B is 1900 g, C is 1008 g and D is 1090 g. In order: 1008, 1080, 1090, 1900. That is C, A, D, B.

4 page 63
Answer: (a) 6 (b) 100
Why: 2 ℓ 500 ml = 2500 ml. 2500 ÷ 400 = 6 remainder 100. He fills 6 cups, and 100 ml are left over.

5 page 64
Answer: (a) 350 (b) 600 (c) 850 (d) 500
Why: Each small mark is 50 ml. A is 1 mark above 300: 350 ml. B is at 600 ml. C is 1 mark above 800: 850 ml. 850 - 350 = 500 ml.

6 page 64
Answer: (a) 4.05 (b) 0.95
Why: (a) 2.65 + 1.40 = 4.05. (b) 5.00 - 4.05 = 0.95. Check: 4.05 + 0.95 = 5.00.

7 page 64
Answer: (a) 14.45 (b) 5.55 (c) Yes
Why: (a) 6.20 + 4.85 = 11.05, and 11.05 + 3.40 = 14.45. (b) 20.00 - 14.45 = 5.55. (c) All 4 things cost 14.45 + 1.75 = 16.20, which is less than $20. So she can.

8 page 64
Answer: 300
Why: 2 kg 250 g = 2250 g. The books weigh 2250 - 450 = 1800 g. One book: 1800 ÷ 6 = 300 g.

9 page 65
Answer: (a) 06:15 (b) 13:40 (c) 21:05 (d) 4:50 (e) 8:25
Why: After 12 noon, add 12 to the hour for 24-hour time. (b) 1 + 12 = 13. (c) 9 + 12 = 21. For (d) and (e), take away 12: 16 - 12 = 4 and 20 - 12 = 8.

10 page 65
Answer: (a) 1 h 15 min (b) 4:35 pm (c) 6:30 pm
Why: (a) 5:25 to 6:00 is 35 min, then 6:00 to 6:40 is 40 min. 35 + 40 = 75 min = 1 h 15 min. (b) 3:40 + 20 min = 4:00, then 35 more min = 4:35 pm. (c) 8:15 pm - 1 h = 7:15 pm, then 45 min back = 6:30 pm.

11 page 65
Answer: (a) 4:10 (b) 7:45 (c) 11:25
Why: The short hand shows the hour and the long hand shows the minutes. A: the long hand points at 2, which is 10 minutes, and the short hand is just past 4. B: the long hand points at 9, which is 45 minutes. C: the long hand points at 5, which is 25 minutes.

12 page 65
Answer: (a) 5.20 (b) 4.80
Why: (a) 8 × 65¢ = 520¢ = $5.20. (b) 10.00 - 5.20 = 4.80.

THINK HARDER

1 page 66
Answer: 07:50
Why: Mei must leave by 08:15 - 25 min = 07:50. The buses leave at 06:50, 07:10, 07:30, 07:50, 08:10. The latest bus that gets her to school in time is the 07:50 bus.

2 page 66
Answer: 3 h 20 min
Why: From 22:50 to 24:00 (midnight) is 1 h 10 min. From midnight to 02:10 is 2 h 10 min. 1 h 10 min + 2 h 10 min = 3 h 20 min.

3 page 66
Answer: (a) 1.20 (b) 2.70
Why: The 2 notebooks cost $3 more than 2 pens in all (2 × $1.50). Take that away: $9 - $3 = $6 for 5 pens. One pen costs $6 ÷ 5 = $1.20. A notebook costs $1.20 + $1.50 = $2.70. Check: 2 × 2.70 + 3 × 1.20 = 5.40 + 3.60 = 9.00.

GO FURTHER

1 page 67
Answer: (a) 2 (b) 4
Why: (a) The 3 ℓ jug takes 3 ℓ, so 5 - 3 = 2 ℓ are left in the 5 ℓ jug. (b) The 3 ℓ jug holds 2 ℓ and has room for 1 ℓ more. The full 5 ℓ jug gives 1 ℓ, so 5 - 1 = 4 ℓ are left. That is exactly 4 ℓ.

2 page 67
Answer: (a) 4:28 pm (b) 4:37 pm
Why: (a) A slow watch is behind: the real time is 8 minutes later: 4:20 pm + 8 min = 4:28 pm. (b) At the real time 4:45 pm her watch shows 8 minutes earlier: 4:37 pm.

3 page 67
Answer: Answers vary. Example: A book costs $6.40 and a pen costs $2.15. How much more does the book cost than the pen? The answer is $4.25.
Why: Check that your story uses both amounts and gives $4.25. The only way is the difference: 6.40 - 2.15 = 4.25. A sum of the two would be $8.55.

4 page 67
Answer: 22:25
Why: From 00:20 go back 20 min to midnight (24:00). 1 h 55 min - 20 min = 1 h 35 min is left. 24:00 - 1 h 35 min = 22:25. Check: 22:25 + 1 h 55 min = 00:20.

CHECK YOURSELF

1 page 68
Answer: 240
Why: 2 m = 200 cm, and 200 + 40 = 240 cm.

2 page 68
Answer: 500
Why: 1 ℓ 200 ml = 1200 ml. Then 1200 - 700 = 500 ml.

3 page 68
Answer: 1 h 25 min
Why: 6:45 to 7:00 is 15 min. 7:00 to 8:00 is 1 h. 8:00 to 8:10 is 10 min. In all: 1 h 25 min.

4 page 68
Answer: 2.65
Why: 5.00 - 2.35 = 2.65. Check: 2.35 + 2.65 = 5.00.

Unit 10 · Puzzle day

Your turn page 69
Answer: 8
Why: 1 child is in front of Siti and 6 are behind her. 1 + 1 + 6 = 8. Check: 2 + 7 - 1 = 8.

START HERE

1 page 70
Answer: (a) Mei **(b)** 15 **(c)** 25 **(d)** 105
Why: (a) The tallest bar is Mei's. (b) 30 - 15 = 15. (c) Siti read 10 books, so Kumar read 10 + 15 = 25. (d) 25 + 15 + 30 + 10 + 25 = 105.

2 page 70
Answer: (a) 70 **(b)** 110 **(c)** 90 **(d)** 270
Why: The labelled lines go up in 20s and there is a thin line halfway between them, so each thin line is 10. A is at 70, B is at 110 and C is at 90. Together: 70 + 110 + 90 = 270.

3 page 70
Answer: (a) Tue **(b)** Mon
Why: (a) From the 3rd to the 24th is 21 days, which is 3 weeks. So the 24th is a Tuesday again. (b) From the 3rd to the 30th is 27 days = 3 weeks and 6 days. 6 days after Tuesday is Monday.

4 page 71
Answer: (a) grapes **(b)** apple **(c)** banana **(d)** orange
Why: Only Lily has a name with 4 letters, so Lily has the orange. Kumar has neither the apple nor the banana, and the orange is taken, so Kumar has the grapes. Wei does not have the apple, so Wei has the banana. Aisha has the apple.

5 page 71
Answer: (a) 9 **(b)** 9
Why: (a) Ravi is 3rd. The 5 children between are 4th to 8th. Mei is 9th and last. (b) 7 children are in front of Aisha and 1 is behind her: 7 + 1 + 1 = 9.

6 page 71
Answer: (a) 369, 396, 639, 693, 936, 963 **(b)** 2 **(c)** 693
Why: Fix the first digit and list the rest in order: 369, 396, then 639, 693, then 936, 963. The even numbers end in 6: 396 and 936. 693 is only 7 from 700. The others are much farther away.

7 page 72
Answer: (a) 31 (b) 8
Why: Each new hexagon adds 5 sticks. The table goes 1: 6, 2: 11, 3: 16, 4: 21, 5: 26, 6: 31. For 41 sticks, count on: 7 hexagons need 36 and 8 hexagons need 41.

8 page 72
Answer: 162, 252 and 342
Why: The hundreds digit is 1, 2 or 3. The ones digit is even and not 0, so it is 2, 4 or 6, and the tens digit is greater. With 1: tens + ones = 8, so 6 and 2. With 2: tens + ones = 7, so 5 and 2. With 3: tens + ones = 6, so 4 and 2. The numbers are 162, 252 and 342.

9 page 72
Answer: 23 and 37
Why: Take away the extra 14: 60 - 14 = 46. Share it into 2 equal parts: 23 each. The numbers are 23 and 23 + 14 = 37. Check: 23 + 37 = 60.

10 page 72
Answer: 2.90
Why: 2 juices cost 2 × 1.85 = 3.70. With the sandwich: 3.70 + 3.40 = 7.10. Change: 10.00 - 7.10 = 2.90.

THINK HARDER

1 page 73
Answer: 21
Why: With 2 friends: 1 handshake. With 3: 3. With 4: 6. Each new friend shakes hands with all the friends already there: 5 friends add 4 (10), 6 add 5 (15) and 7 add 6 (21).

2 page 73
Answer: Aisha
Why: If Ben did it: Aisha tells the truth, and Wei tells the truth (Ben's words are a lie). That is 2 true statements. If Wei did it: Ben and Aisha both tell the truth. If Aisha did it: Ben lies, Aisha lies and Wei tells the truth. That is exactly 1. So Aisha did it.

3 page 73
Answer: (a) 30 (b) 45 (c) 60 (d) 50
Why: Monday is 1 box. Tuesday is 1 box and 15. Wednesday is 2 boxes. Thursday is 2 boxes take away 10. In all: 6 boxes + 15 - 10 = 6 boxes + 5 = 185. So 6 boxes = 180 and 1 box = 30. Monday 30, Tuesday 45, Wednesday 60, Thursday 50. Check: 30 + 45 + 60 + 50 = 185.

GO FURTHER

1 page 74
Answer: (a) 16 (b) 9 (c) 4 (d) 30
Why: There are 16 small squares. A 2 by 2 square fits in 3 places across and 3 down: 9. A 3 by 3 square fits in 2 places across and 2 down: 4. There is 1 big square. 16 + 9 + 4 + 1 = 30.

2 page 74
Answer: (a) 1 (b) 2
Why: (a) Put 1 coin on each side. If the scale tips, the light coin is on the high side. If it balances, it is the coin left out. That is 1 weighing. (b) Weigh 3 coins against 3 coins. This shows which group of 3 has the light coin. Then use the 1 weighing from (a). That is 2 weighings.

3
page 74

Answer: (a) 1 (b) 7
Why: If all 8 coins were 10¢, they would be worth 80¢. They are worth 150¢, which is 70¢ more. Changing one 10¢ coin to a 20¢ coin adds 10¢, so 7 coins are 20¢ and 1 coin is 10¢. Check: 10 + 7 × 20 = 150.

4
page 74

Answer: Answers vary. Example: Lily, Wei and Kumar each play a different game: chess, ball or cards. Lily does not play ball. Wei plays neither chess nor cards. Kumar does not play chess. Answer: Lily plays chess, Wei plays ball, Kumar plays cards.
Why: Check that your clues are true for your answer and that every other arrangement breaks a clue. In the example, Wei plays ball (clue 2), then Kumar cannot play chess (clue 3), so Kumar plays cards and Lily plays chess.

CHECK YOURSELF

1
page 75

Answer: 15
Why: Monday is at 20 and Tuesday is at 35. 35 - 20 = 15.

2
page 75

Answer: 8
Why: 3 + 1 + 4 = 8. Do not forget Ben.

3
page 75

Answer: 2
Why: The numbers are 147, 174, 417, 471, 714 and 741. The even ones end in 4: 174 and 714.

4
page 75

Answer: 30
Why: The numbers go up by 4. The 8th number is 2 + 7 × 4 = 30.